

Culminating Activity

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May 16, 2026

Abstract

This study examines the determinants of software developer compensation using the Developer Survey 2026 dataset, a synthetically generated dataset comprising 12,000 records and 32 variables. Employing multiple linear regression (OLS) as the primary analytical method, the study models log-transformed annual salary as a function of 16 predictor groups encompassing developer demographics, employment characteristics, technical roles, educational attainment, geographic location, and artificial intelligence-related behavioral variables. Six model specifications were estimated and compared using Adjusted R^2 , AIC, and BIC criteria. The full model achieved an Adjusted R^2 of 0.9218, indicating that approximately 92.18% of the variance in log-transformed developer salary is explained by the included predictors ($F(102, 11342) = 1,324, p < 2.2e-16$). Results indicate that years of experience, employment status, developer role, education level, and country of employment are the most statistically significant predictors of compensation, while AI-related variables including productivity boost percentage, usage frequency, and sentiment did not contribute independently to salary variation once structural factors were controlled. Comprehensive OLS assumption diagnostics confirmed that the model satisfies the conditions of homoscedasticity, error independence, and approximate normality of residuals. These findings suggest that structural and human capital factors remain the dominant drivers of developer compensation, with geographic context exerting the largest categorical influence. The study contributes a rigorous regression-based framework for analyzing workforce compensation in the software development industry and offers data-driven implications for developers, employers, academic institutions, and labor market researchers.

Chapter 1: Introduction

Background of the Study

The global demand for software developers continues to rise as organizations accelerate digital transformation and adopt artificial intelligence (AI) in their operations. Industry outlooks show that software development remains a high-growth field, with employment projected to expand much faster than average and with strong median wages that reflect sustained labor demand (U.S. Bureau of Labor Statistics, 2025). This makes it important to examine the factors that shape developer compensation, skill requirements, and career opportunities.

Recent studies also indicate that AI is not only changing how software is built but also reshaping the role of developers. Morgan Stanley Research reported that AI coding tools are expected to enhance productivity and may increase hiring demand as enterprises build more complex applications and address technical debt (Morgan Stanley, 2025). In the same direction, Forbes noted that AI is transforming the software developer role by automating routine coding tasks and shifting developers toward more strategic work such as architecture, integration, and product oversight (Drenik, 2024).

In addition, salary trends suggest that experience and specialized technical skills remain major determinants of developer compensation. Dice reported that software developer salaries continue to increase, with higher pay associated with greater experience and stronger demand for modern technical skills (Dice, 2025). Likewise, digital transformation continues to require developers to expand their competencies in areas such as cloud computing, AI, machine learning, and DevOps, reinforcing the need to understand how skills and compensation are related in the evolving labor market (RVS Media, 2025).

Given these developments, analyzing the Developer Survey 2026 dataset can provide useful insights into developer skills, salary levels, experience, and perceptions of AI (Stack Overflow, 2026). To systematically examine these relationships, this study will apply a machine learning approach using a multilinear regression model, which enables the analysis of how multiple independent variables, such as years of experience, skill sets, and AI usage, simultaneously influence a dependent variable, particularly developer salary (Suman, 2025; Putra et al., 2026). By leveraging multilinear regression, the study aims to quantify the strength and direction of these relationships, identify significant predictors of compensation, and generate data-driven insights into how evolving technologies and skill demands impact the software development workforce (García et al., 2025; Rodrigues, 2022).

Statement of the Problem

The rapid evolution of technology, particularly the rise of artificial intelligence, has significantly influenced the software development industry. Developers are now expected to continuously adapt by acquiring new skills, while organizations adjust compensation based on experience, expertise, and emerging technological demands. Despite the availability of large-scale survey data, there is still a need to clearly understand how these factors interact and influence developer salaries and roles using quantitative and predictive approaches.

In particular, traditional descriptive analysis alone may not fully capture the combined effect of multiple variables on salary. Thus, there is a need to apply a multilinear regression model to examine how several independent variables, such as skills, experience, and AI impact, simultaneously influence developer compensation.

This study aims to analyze the Developer Survey 2026 dataset to identify key patterns and relationships among skills, experience, salary, and the impact of artificial intelligence. Specifically, it seeks to answer the following research questions:

1. What is the relationship between developers' skills, experience, and salary based on the Developer Survey 2026 dataset using a multilinear regression model?
2. How does the impact of artificial intelligence influence developers' roles and salary levels when analyzed through a multilinear regression approach?

Objectives of the Study

General Objective

To comprehensively investigate the relationships among developer skills, experience, salary, and the impact of artificial intelligence using the Developer Survey 2026 dataset through the application of a multilinear regression model and other data analysis techniques.

Specific Objectives

- To determine the extent to which years of experience influence developer salary using regression coefficients.
- To analyze how different programming languages, tools, and technical skills contribute to variations in developer compensation within a multilinear regression framework.
- To identify which developer roles or specializations significantly predict salary levels.
- To examine how developers perceive the impact of artificial intelligence on their productivity, job responsibilities, and career opportunities.
- To evaluate whether AI-related variables significantly affect salary and roles based on regression results.
- To develop and interpret a multilinear regression model that explains salary as a function of multiple independent variables.
- To assess the strength and significance of relationships using statistical measures such as coefficients, p-values, and R^2 .

Hypotheses

Hypothesis	Statement
$H0_1$	The multilinear regression model shows that developers' experience does not significantly predict salary.
$H1_1$	The multilinear regression model shows that developers' experience significantly predicts salary.
$H0_2$	The multilinear regression model shows that developers' technical skills do not significantly predict salary.
$H1_2$	The multilinear regression model shows that developers' technical skills significantly predict salary.
$H0_3$	The multilinear regression model shows that artificial intelligence has no significant effect on developers' roles and salary.
$H1_3$	The multilinear regression model shows that artificial intelligence has a significant effect on developers' roles and salary.

Significance of the Study

This study is significant as it applies a multilinear regression model to provide a more precise and data-driven understanding of the factors influencing developer salaries and career development. Instead of relying solely on descriptive insights, the study quantifies the impact of multiple variables simultaneously.

- **Students and Aspiring Developers:** Gain insight into which specific skills and factors statistically influence salary, helping them prioritize learning areas.
- **Professional Developers:** Understand how different variables contribute to compensation, enabling more informed career decisions.
- **Employers and Recruiters:** Benefit from predictive insights that help in structuring competitive salary packages based on measurable factors.
- **Academic Institutions:** Can incorporate data-driven findings into curriculum design, especially in AI and in-demand technologies.
- **Researchers:** Provides a methodological contribution by demonstrating the application of multilinear regression in analyzing workforce and salary data.

Scope and Limitations

Scope

This study focuses on the analysis of the Developer Survey 2026 dataset, which includes variables related to developer demographics, technical skills, experience, salary, and perceptions of artificial intelligence.

The research specifically applies a multilinear regression model to examine how multiple independent variables (such as skills, experience, and AI-related factors) simultaneously influence the dependent variable (salary). Additional techniques such as descriptive statistics and correlation analysis are also used to support the findings.

The study aims to produce both explanatory and predictive insights regarding salary determination in the software development industry.

Limitations

The study is limited by the quality and structure of the dataset, which is based on self-reported survey responses and may include biases or inconsistencies. Additionally, the regression model assumes linear relationships between variables, which may not fully capture complex real-world interactions.

Some potentially important variables, such as geographic location, company size, or economic conditions, may not be fully represented, which can affect the accuracy of the model. Furthermore, while multilinear regression can identify significant predictors, it does not establish causation, and results should be interpreted accordingly.

Chapter 2: Review of Related Literature

Related Literature

The rapid adoption of artificial intelligence into software development has fundamentally changed the industry, introducing both significant opportunities and complex challenges (Garg, 2023). AI-powered tools have reshaped conventional development practices by automating key phases of the software development lifecycle, including code writing, testing and debugging, and deployment (Madupati, 2025). These shifts have not only altered how software is built in the software industry, but have also redefined the skills and expertise that employers prioritize, which influences how developers are compensated. As AI continues to permeate the software industry, understanding the factors that drive compensation has become increasingly relevant for both practitioners and researchers alike.

In the fast-paced software industry, salary has emerged as a critical factor in workforce dynamics. Research has established that employee experience is positively associated with motivation, largely driven by better compensation (Memon et al., 2021). This relationship suggests that salary is not merely a financial metric but also a signal of professional value and career progression, particularly in technically demanding fields such as software and AI development. Furthermore, one study found a strong pull of AI talent in the labor market that challenges narratives of technological displacement, highlighting instead how employers actively compete for scarce AI talent through both financial and non-financial incentives (Castaneda et al., 2025). This intensifying competition for skilled professionals further underscores the importance of identifying which factors most significantly drive compensation in the AI and software development sectors.

To rigorously examine such relationships, appropriate analytical methods are essential — and several literatures examine the chosen methods the methodology makes use of. Variable selection was conducted to retain only the most theoretically and statistically relevant predictors, as retaining unnecessary variables risks biased coefficient estimates and reduced model interpretability (Hanke et al., 2024). To evaluate model performance reliably, cross-validation was employed, as it provides a more robust estimate of predictive accuracy than a single train-test evaluation alone (Bates et al., 2024). The dataset was partitioned into training and testing subsets following standard practice, with the choice of split strategy informed by its demonstrated impact on model generalizability (Bami et al., 2025). Multiple linear regression was selected as the primary analytical method given its interpretability, its suitability for continuous outcome variables, and its continued validity as a methodology for salary prediction research (Tappari et al., 2026; Keith et al., 2025).

Related Studies

Prior studies have explored salary prediction using a wide range of machine learning and statistical approaches, yielding valuable methodological and substantive insights. A comprehensive literature survey examining salary prediction across multiple domains and datasets

found that multiple linear regression remains a valid, interpretable, and widely used methodology, even as more advanced algorithms have gained traction in recent years (Tappari et al., 2026). The continued relevance of linear regression in this space speaks to its strength in producing transparent models that clearly communicate the directional and magnitude effects of individual predictors, a quality that is particularly valuable in applied research where interpretability matters. That said, research has also noted persistent challenges in technicality and model reusability when combining multiple linear regression with other machine learning approaches, suggesting that while hybrid models may boost predictive accuracy, they can introduce complexity that limits practical applicability (Kumar et al., 2023).

When it comes to identifying key salary determinants, previous regression-based studies consistently point to skills, experience, and location as the most influential factors shaping compensation outcomes. A multiple linear regression model with an R^2 of 0.82 found that location and experience were the most statistically and practically significant predictors of compensation in the AI industry, demonstrating that geographic context and years of experience together account for a substantial proportion of salary variance (Maidin et al., 2024). These findings align with broader labor market theory, which holds that compensation reflects both the scarcity of skills in a given market and the accumulated human capital of individual workers.

A more recent study by Godase (2026), employing random forest regression on a dataset of over 10,000 job postings and achieving an R^2 of 0.98, corroborated these findings and concluded that experience, technical skills, and organizational characteristics are crucial determinants of compensation. The high model fit achieved in that study further validates the importance of these variables, while the use of a large-scale job posting dataset adds real-world credibility to its findings. Further supporting this, Xu (2025) found that salary growth tends to accelerate at higher experience levels, suggesting that the financial returns to experience are not uniform but increase disproportionately as professionals advance in their careers. This has important implications for workforce planning and retention strategies in the AI and software sectors.

Deb (2025) similarly found that job title, experience level, work arrangement, employee residence, and company size all significantly contribute to salary variation, with full-time employment, on-site work, U.S. residency, and medium-sized companies consistently associated with higher salaries. This points to the multidimensional nature of compensation, where structural and organizational factors interact with individual-level attributes to determine pay outcomes. Taken together, these studies paint a consistent picture: compensation in the software and AI industry is shaped by a combination of individual human capital factors, particularly experience and technical skills, and contextual variables such as location, company characteristics, and employment arrangement.

Chapter 3: Methodology

3.1 Research Design

This study adopts a quantitative, non-experimental research design using secondary data obtained from the Kaggle Developer Survey dataset. The research is descriptive and predictive in nature, aiming to identify factors that significantly influence developer compensation and to construct a reliable predictive model for annual salary.

The analytical approach is grounded in multiple linear regression, a classical statistical method suited for estimating the linear relationship between a continuous dependent variable and multiple independent predictors. Because the data are observational rather than experimentally controlled, the study focuses on identifying statistically significant associations rather than making causal claims.

Model validity is assessed through a comprehensive suite of diagnostic tests covering the core OLS assumptions: normality of residuals, homoscedasticity, independence of errors, and absence of severe multicollinearity. Multiple model specifications are compared using Adjusted R^2 , AIC, and BIC to arrive at the most parsimonious and well-fitting final model.

3.2 Data Collection

The dataset used in this study was sourced from Kaggle, a publicly accessible platform for open datasets widely used in academic and data science research. The dataset is a **synthetically generated** developer survey containing 12,000 records and 32 variables designed to simulate realistic developer demographics, compensation, and technology adoption patterns.

Dataset Source: <https://www.kaggle.com/datasets/meruvakodandasuraj/developer-survey-2026-skills-salary-and-ai-impact>

The dataset was loaded into R using the following command:

Table 2: Table 1. Developer Demographics

respondent_id	survey_year	country	age	gender	education_level	years_of_experience	role
R00001	2024	India	43	Man	Some college	22	Front-End Developer
R00002	2025	Italy	28	Man	Master's degree	7	Machine Learning Engineer
R00003	2021	South Africa	23	Woman	Master's degree	0	Machine Learning Engineer
R00004	2023	India	22	Prefer not to say	Bachelor's degree	1	QA / Test Engineer
R00005	2025	United States	34	Man	Bachelor's degree	11	Front-End Developer
R00006	2021	India	33	Man	Bachelor's degree	15	Front-End Developer
R00007	2022	China	29	Man	Bachelor's degree	13	Cloud Engineer
R00008	2022	India	31	Man	Bachelor's degree	12	Full-Stack Developer
R00009	2025	United States	46	Man	Bachelor's degree	26	Front-End Developer
R00010	2020	Indonesia	37	Man	Self-taught / Bootcamp	11	Front-End Developer
R00011	2021	Pakistan	20	Prefer not to say	Master's degree	0	Embedded Developer
R00012	2026	United States	36	Woman	Master's degree	16	QA / Test Engineer
R00013	2022	South Africa	50	Man	Self-taught / Bootcamp	25	Full-Stack Developer
R00014	2025	Belgium	24	Man	Bachelor's degree	0	Front-End Developer
R00015	2021	India	21	Man	Master's degree	2	Student / Intern

Table 3: Table 2. Employment and Salary Information

employment_status	company_size	work_mode	coding_hours_per_week	annual_salary_usd	job_satisfaction	career_satisfaction
Unemployed	1-10	Hybrid	12	3700	1	4
Unemployed	201-1000	Remote	26	9660	3	4
Full-time	1001-5000	In-Office	43	26200	4	2
Unemployed	10000+	Hybrid	27	0	5	2
Full-time	51-200	Remote	25	172100	3	4
Full-time	51-200	In-Office	29	14900	4	4
Full-time	1001-5000	Hybrid	27	45200	1	5
Full-time	51-200	Remote	34	16200	5	4
Full-time	201-1000	Hybrid	40	245100	4	5
Full-time	1001-5000	In-Office	45	17900	2	4
Student	5001-10000	Hybrid	12	0	4	3
Full-time	11-50	In-Office	33	211500	3	4
Full-time	1-10	Hybrid	35	46500	5	4
Full-time	5001-10000	Remote	41	34600	4	4
Full-time	201-1000	Hybrid	51	4600	4	5

Table 4: Table 3. Technology Stack

primary_os	languages_used	frameworks_used	databases_used	cloud_platforms_used	ide_editor
macOS	Swift;JavaScript;Python;HTML/CSS;SQL;Kotlin	Node.js;Ruby on Rails	PostgreSQL;Firebase Realtime DB;SQLite;Neo4j	AWS;Cloudflare;Hetzner	Zed
macOS	Python;HTML/CSS;Java	FastAPI	PostgreSQL;SQLite;MongoDB;MySQL	Cloudflare	Zed;Visual Studio;IntelliJ IDEA
Linux	Swift;Python;Java;JavaScript;Bash/Shell;SQL	Spring Boot	DynamoDB	Google Cloud;Railway	VS Code;Visual Studio
Linux	Python;JavaScript;TypeScript	Express	PostgreSQL;Oracle;Supabase	DigitalOcean;Netlify;Vercel	Vim/Neovim;VS Code
Windows	C++;JavaScript;Scala;TypeScript	Django;.NET;Vue.js;Next.js;React	Firebase Realtime DB;SQLite;Redis;ClickHouse	Cloudflare;Vercel	VS Code;Vim/Neovim;Visual Studio
Windows	JavaScript;C;Python;SQL	React;Node.js;Laravel;Express;Flutter	SQLite;Microsoft SQL Server;PostgreSQL	Heroku	WebStorm;Visual Studio
Linux	PHP;Bash/Shell;SQL;Python	Spring Boot;htmx;Tailwind CSS;Bootstrap	MySQL	AWS;Azure	VS Code;Cursor
Windows	C;Kotlin;Bash/Shell;SQL;JavaScript	FastAPI;React	Microsoft SQL Server	AWS;Azure	VS Code;Vim/Neovim
macOS	JavaScript	Flask;Next.js;TensorFlow	Redis;PostgreSQL;Supabase;MySQL	AWS	Visual Studio;IntelliJ IDEA;VS Code
Windows	C#;Rust;HTML/CSS	Vue.js;Bootstrap	DynamoDB;PostgreSQL;CockroachDB;MariaDB	DigitalOcean;Vercel	Android Studio;Xcode;VS Code
macOS	SQL;JavaScript	Django;Flask;Angular;Next.js	PostgreSQL;Microsoft SQL Server	Azure;DigitalOcean	WebStorm;PyCharm;Vim/Neovim
macOS	SQL;Go;Python;JavaScript	TensorFlow;Tailwind CSS;NestJS	Firebase Realtime DB;PostgreSQL	Fly.io	Sublime Text;IntelliJ IDEA
Windows	PHP;HTML/CSS;TypeScript;SQL	Spring Boot;Next.js	Cassandra;Neo4j;MySQL;PostgreSQL	AWS	VS Code;IntelliJ IDEA;Cursor
Windows	Zig;Python;JavaScript;C++	Angular	SQLite;PostgreSQL	Google Cloud;Azure;Netlify	IntelliJ IDEA;Android Studio
Windows	PHP	Node.js;jQuery;FastAPI;Spring Boot	Microsoft SQL Server;Redis;SQLite;DynamoDB	Cloudflare	VS Code;Cursor;Vim/Neovim

Table 5: Table 4. AI Usage and Perception

ai_tools_used	ai_usage_frequency	ai_productivity_boost_pct	ai_sentiment	ai_job_threat_perception
ChatGPT	Weekly	14	Somewhat positive	Minimal threat
	Weekly	8	Somewhat positive	Minimal threat
	Rarely	1	Very negative	Significant threat
	Multiple times daily	61	Neutral	Minimal threat
	Weekly	15	Neutral	Moderate threat
	Daily	36	Very positive	Significant threat
ChatGPT	Monthly	8	Somewhat positive	Minimal threat
	Multiple times daily	16	Neutral	No threat
	Weekly	26	Somewhat positive	Will replace most devs
ChatGPT;Cursor AI	Weekly	19	Very negative	Significant threat
Cursor AI;ChatGPT;Claude	Never	9	Somewhat positive	Moderate threat
ChatGPT	Multiple times daily	36	Very positive	No threat
	Monthly	4	Very positive	Minimal threat
	Daily	7	Very positive	Moderate threat
Claude Code;Perplexity	Multiple times daily	45	Very positive	Minimal threat

An initial inspection confirmed that only one column, `ai_tools_used`, contained missing values (approximately 19.8% of rows). All other columns were complete prior to imputation.

Table 6: Table 5. Career Growth and Preferences

primary_learning_source	open_source_contributor	looking_for_job	remote_work_preference_1to5	burnout_frequency	want_to_learn_next
Online courses	1	Actively looking	2	Sometimes	Elixir;Zig;AI/ML
Stack Overflow	1	Not looking	4	Often	Kubernetes;Elixir;Go
YouTube	0	Open to offers	5	Often	Go
Stack Overflow	1	Open to offers	3	Rarely	HTMX;Swift;Rust
Official docs	0	Open to offers	1	Sometimes	TypeScript
Stack Overflow	0	Not looking	3	Sometimes	Kubernetes;TypeScript
Official docs	0	Not looking	5	Often	AI/ML;Go
Books	1	Not looking	5	Often	Kotlin;Swift;Kubernetes
YouTube	0	Not looking	5	Often	Swift;HTMX
ChatGPT/AI	1	Actively looking	4	Rarely	Rust;Kotlin;Go
Colleagues	0	Not looking	2	Always	Rust
Stack Overflow	0	Not looking	5	Rarely	Elixir;Zig;HTMX
Blogs	1	Open to offers	3	Always	Go;Elixir
ChatGPT/AI	1	Open to offers	4	Sometimes	TypeScript;Swift
Official docs	0	Open to offers	4	Often	Python;HTMX

The `ai_tools_used` column was excluded from the model as it is a semicolon-delimited multi-value free-text field not directly suitable for regression encoding.

3.3 Variables Description

The study models annual developer salary as the dependent variable, with a comprehensive set of demographic, employment, educational, and behavioral predictors serving as independent variables.

3.3.1 Dependent Variable

The dependent variable is `log_salary`, defined as the natural logarithm of `annual_salary_usd`. A log transformation was applied because raw salary distributions are strongly right-skewed; the transformation stabilizes variance across fitted values and brings the distribution closer to normality, both of which are assumptions of OLS regression. Respondents with `annual_salary_usd` ≤ 0 were excluded from the analysis as these entries reflect unemployed or unreported salary statuses rather than true compensation values.

3.3.2 Independent Variables

The independent variables include all numeric, ordinal, and nominal features available in the cleaned dataset after excluding ID columns, multi-value text fields, and the target variable. Table 3.1 presents the full set of predictors, their measurement type, and how each was encoded for regression.

Variable	Type	Encoding / Notes
years_of_experience	Continuous numeric	Raw numeric; direct linear predictor of salary
age	Continuous numeric	Raw numeric
coding_hours_per_week	Continuous numeric	Raw numeric
ai_productivity_boost_pct	Continuous numeric	Raw numeric; self-reported % boost from AI tools
remote_work_preference_1to5	Ordinal numeric	Raw numeric (1 = fully on-site, 5 = fully remote)
open_source_contributor	Binary numeric	Raw numeric (0/1)
education_level	Ordered factor	7-level ordered factor from High school to Doctorate (PhD)
company_size	Ordered factor	7-level ordered factor from 1–10 to 10000+ employees
burnout_frequency	Ordered factor	5-level ordered factor: Never to Always
ai_usage_frequency	Ordered factor	6-level ordered factor: Never to Multiple times daily
role	Nominal factor	Unordered; each level gets a dummy coefficient vs. baseline
employment_status	Nominal factor	Unordered; e.g., Full-time, Part-time, Freelance
work_mode	Nominal factor	Unordered; e.g., Remote, Hybrid, In-Office
gender	Nominal factor	Unordered
primary_os	Nominal factor	Unordered; e.g., Windows, macOS, Linux
ai_sentiment	Nominal factor	Unordered; attitude toward AI in the workplace
ai_job_threat_perception	Nominal factor	Unordered; perceived job threat level from AI
primary_learning_source	Nominal factor	Unordered; e.g., Official docs, Stack Overflow, YouTube
looking_for_job	Nominal factor	Unordered; current job-seeking status

Variable	Type	Encoding / Notes
country	Nominal factor	Unordered; country of employment

Table 3.1. Independent Variables Used in the Multiple Linear Regression Model

3.4 Data Analysis Techniques

3.4.1 Descriptive Statistics

Exploratory data analysis was conducted prior to modeling to understand variable distributions and guide preprocessing decisions. Specific procedures included:

- Histograms of raw and log-transformed salary to justify the log transformation
- Pearson correlation matrix of all numeric predictors against `log_salary` to screen for linear relationships
- Summary statistics (mean, median, standard deviation, quartiles) via `summary()` in R
- Frequency tables for categorical variables to detect dominant categories and inform mode imputation
- Missing value audit to quantify and locate all NA entries before imputation

Numeric variables with missing values were imputed using the column median, and categorical variables were imputed using the column mode, both strategies chosen to preserve distributional properties without introducing bias from mean substitution in skewed data.

3.4.2 Inferential Statistics

The primary analytical method is multiple linear regression (MLR) estimated by Ordinary Least Squares (OLS), implemented using the `lm()` function in R. The regression equation takes the general form:

$$\log(\text{salary}) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \varepsilon$$

where X_1 through X_k represent the independent variables listed in Table 3.1, β_0 is the intercept, $\beta_1 \dots \beta_k$ are regression coefficients, and ε is the error term. All hypothesis tests are conducted at the $\alpha = 0.05$ significance level.

Significance Testing

- Individual predictor significance was assessed via t-tests on each regression coefficient, with p-values reported from `summary(model)`

- Overall predictor group significance was assessed via F-tests using `anova(model)`
- Predictors were classified as significant ($p < 0.05$) or non-significant ($p \geq 0.05$) and ranked accordingly

Variable Selection

To identify the most parsimonious model, several variable selection strategies were applied and compared:

- Manual stepwise elimination by p-value, producing Models v2, v3, and v4 by progressively removing non-significant predictors
- Backward elimination using `drop1()` with F-tests to identify the weakest contributor at each step
- Bidirectional stepwise AIC selection using `stepAIC(direction = "both")` from the MASS package
- Stepwise BIC selection using `stepAIC()` with `k = log(n)` to apply a stronger complexity penalty
- A theory-driven reduced model retaining only substantively meaningful predictors

All candidate models were evaluated and ranked on Adjusted R^2 , AIC, and BIC. The full model was retained as the final model based on this comparison.

Model Assumption Diagnostics

The following formal tests and visual diagnostics were applied to the final model to validate the OLS assumptions:

Assumption	Test / Method	Tool in R
Normality of residuals	Shapiro-Wilk test; Kolmogorov-Smirnov test; Q-Q plot; residual histogram	<code>shapiro.test()</code> , <code>ks.test()</code> , <code>qqnorm()</code>
Homoscedasticity	Breusch-Pagan test; Residuals vs. Fitted plot; Scale-Location plot	<code>bptest()</code> from <code>lmtest</code>
Independence of errors	Durbin-Watson test (acceptable range: 1.5 – 2.5)	<code>dwtest()</code> from <code>lmtest</code>
Multicollinearity	Variance Inflation Factor (VIF); flag $VIF > 5$ (moderate) and $VIF > 10$ (severe)	<code>vif()</code> from <code>car</code>
Influential observations	Cook's Distance with threshold $4/n$; Leverage (hat values) with threshold $2p/n$	<code>cooks.distance()</code> , <code>hatvalues()</code>

Table 3.2. OLS Assumption Diagnostics Applied to the Final Model

In summary, this chapter described a rigorous multiple linear regression pipeline implemented in R, covering data sourcing from Kaggle, preprocessing and imputation, variable operationalization, model fitting, variable selection across multiple strategies, and a comprehensive set of assumption diagnostics. The combination of formal statistical tests and visual diagnostics ensures that the final model is both statistically valid and interpretable.

Chapter 4: Results and Discussions

This chapter presents the results of the multiple linear regression analysis conducted on the Developer Survey 2026 dataset. The findings are organized into four sections: descriptive statistics summarizing the dataset, visualizations of key variable distributions and relationships, model outputs from all candidate regression specifications, and a discussion interpreting the significant predictors and diagnostic results.

4.1 Descriptive Statistics

Descriptive statistics were computed on the cleaned dataset (`data_clean`) using the `summary()` function in R. After removing rows with zero or missing salary values, the final dataset consisted of 11,445 observations across all retained variables.

Table 4.1 presents the descriptive statistics for the continuous variables in the cleaned dataset.

Variable	Min	Median	Mean	Max
survey_year	2020	2024	2023	2026
age	18.0	31.0	31.3	60.0
years_of_experience	0.000	8.000	9.389	41.000
coding_hours_per_week	5.00	33.00	32.89	67.00
annual_salary_usd	2,000	38,000	58,625	368,700
job_satisfaction	1.000	4.000	3.602	5.000
career_satisfaction	1.000	4.000	3.748	5.000
ai_productivity_boost_pct	0.00	23.00	24.46	80.00
open_source_contributor	0.000	0.000	0.345	1.000
remote_work_preference_1to5	1.000	4.000	4.056	5.000
log_salary	7.601	10.545	10.509	12.818

Table 4.1a. Descriptive Statistics — Continuous Variables

Variable	Most Frequent	Count	2nd Most Frequent	Count
country	United States	2,067	India	1,611
gender	Man	10,025	Woman	970
education_level	Bachelor's degree	4,878	Master's degree	3,194

Variable	Most Frequent	Count	2nd Most Frequent	Count
role	Full-Stack Developer	2,564	Back-End Developer	1,701
employment_status	Full-time	9,325	Freelance	1,008
company_size	201-1000	2,233	51-200	2,126
work_mode	Hybrid	4,597	Remote	4,406
primary_os	Windows	4,761	macOS	3,706
ai_usage_frequency	Multiple times daily	3,154	Daily	2,868
ai_sentiment	Somewhat positive	4,063	Neutral	2,553
ai_job_threat_perception	Minimal threat	3,410	Moderate threat	3,221
primary_learning_source	Official docs	2,292	Stack Overflow	2,058
looking_for_job	Open to offers	4,549	Not looking	4,351
burnout_frequency	Sometimes	3,999	Often	2,906

Table 4.1b. Descriptive Statistics — Categorical Variables (Top 2 Categories)

4.2 Visualizations

Visual exploration of the data was conducted using `ggplot2` in R. The following plots were generated to understand the distributions of key variables and their relationships with the log-transformed salary outcome.

4.2.1 Salary Distribution

The raw salary distribution exhibited strong right skewness, with a long upper tail driven by high-earning roles and senior developers. After applying the natural log transformation, the distribution became approximately symmetric and bell-shaped, confirming that the log transformation was appropriate for meeting the normality assumptions of OLS regression.

4.2.2 Log Salary by Role

Figure 4.2 presents the distribution of log-transformed salary across developer roles. Engineering Manager and DevOps/SRE consistently occupy the highest salary range, with median log salaries approaching 11.0 and above, reflecting strong market demand for infrastructure and leadership roles. Security Engineer and Machine Learning Engineer follow

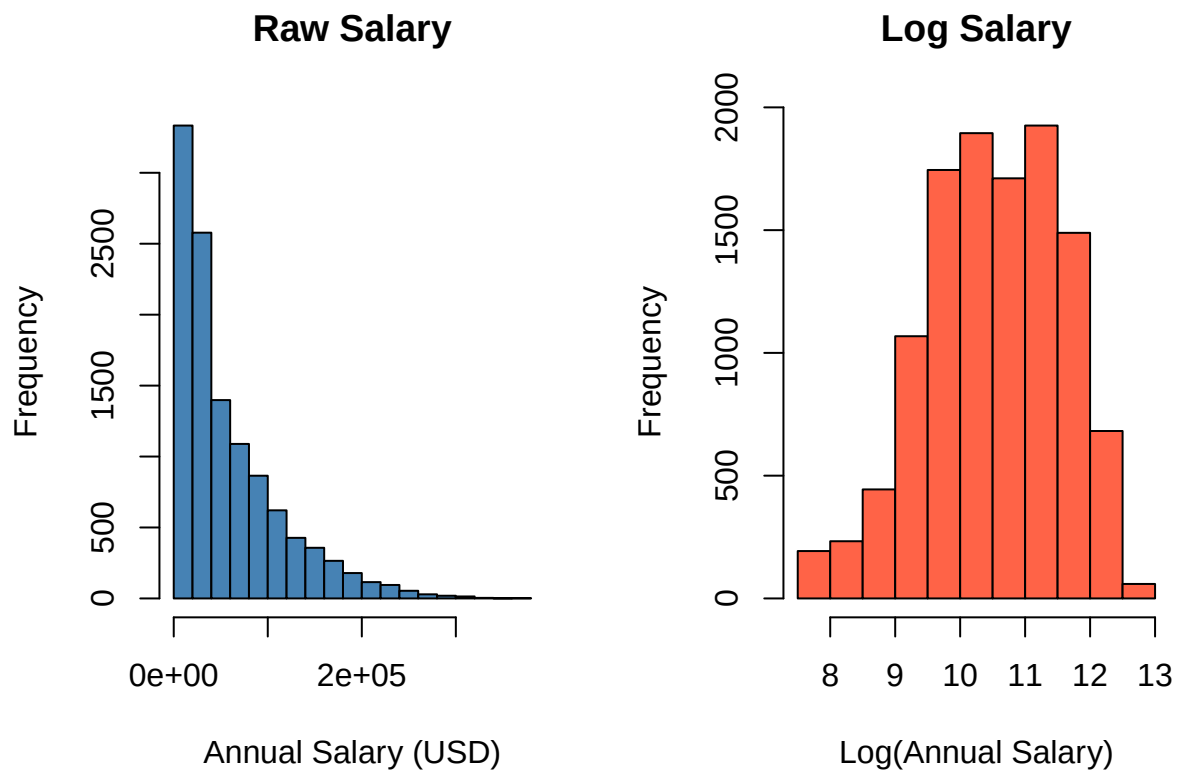


Figure 1: Figure 4.1. Raw vs. Log-Transformed Salary Distribution

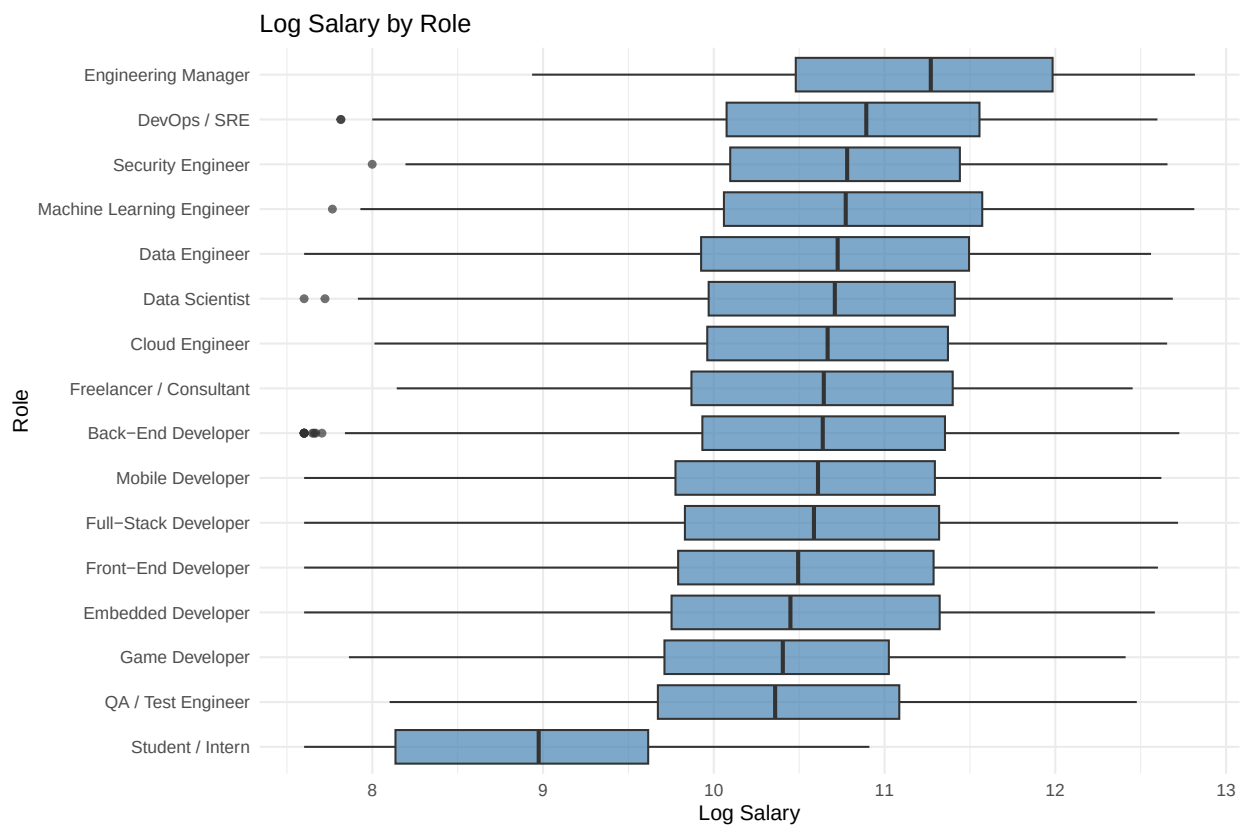


Figure 2: Figure 4.2. Log Salary by Developer Role

closely, both showing relatively compact interquartile ranges suggesting more consistent compensation within those roles.

Mid-tier roles such as Data Engineer, Data Scientist, Cloud Engineer, and Freelancer/Consultant cluster around a median log salary of approximately 10.5, aligning with the overall dataset median. Back-End, Mobile, Full-Stack, and Front-End Developers occupy a similar range but with wider spreads, indicating greater variability in compensation likely driven by geography, company size, and experience level.

The lowest salary distributions belong to Game Developer, Embedded Developer, QA/Test Engineer, and notably Student/Intern — the latter showing a distinctly lower median of approximately 8.5 and a compressed range consistent with entry-level or unpaid compensation structures. The clear separation between Student/Intern and all other roles confirms that employment status and role type are among the most influential structural determinants of developer salary, which is further supported by their statistical significance in the regression model.

4.2.3 Log Salary by Employment Status

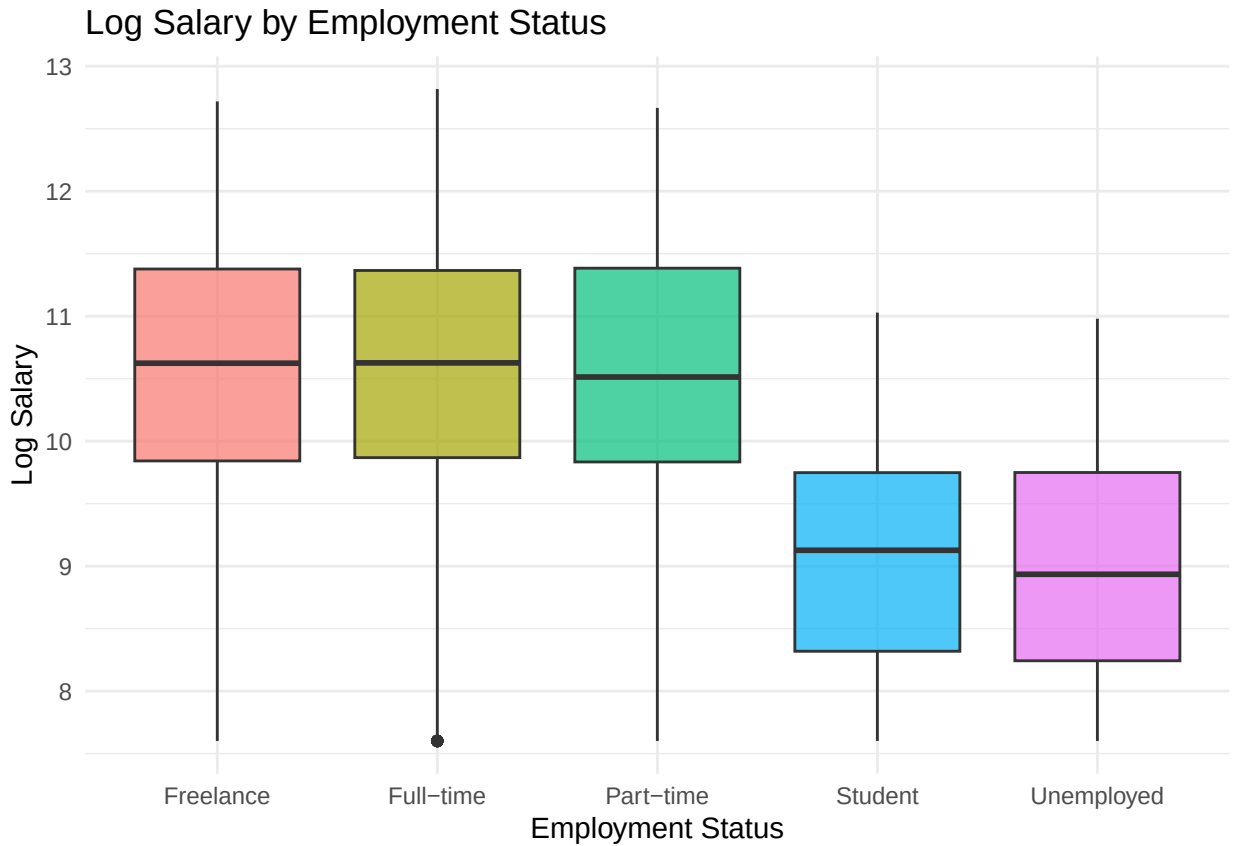


Figure 3: Figure 4.3. Log Salary by Employment Status

Figure 4.3 displays the distribution of log-transformed salary across the five employment sta-

tus categories. Freelance, Full-time, and Part-time developers share nearly identical median log salaries of approximately 10.7–10.8, with similarly wide interquartile ranges spanning roughly 10.0 to 11.5, suggesting that compensation levels are broadly comparable across these active employment types. In stark contrast, Student and Unemployed respondents show substantially lower median log salaries of approximately 9.0, with compressed distributions and notably less upward spread, reflecting the structural absence of full professional compensation in these categories. The single outlier visible below the Full-time whisker indicates a small number of full-time developers reporting unusually low salaries. The sharp divide between the three active employment categories and the Student/Unemployed groups reinforces the regression finding that employment status is one of the most statistically significant predictors of developer salary.

4.2.4 Log Salary vs. Years of Experience

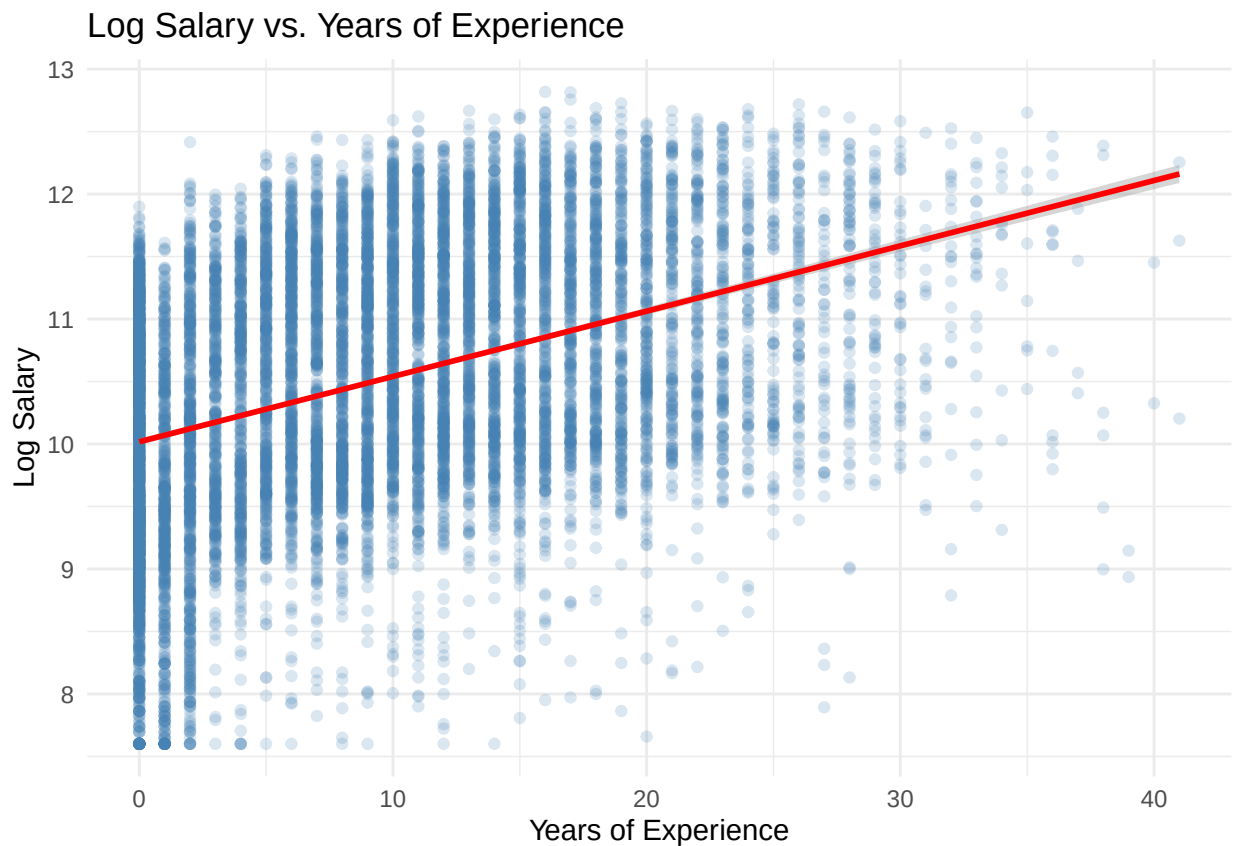


Figure 4: Figure 4.4. Log Salary vs. Years of Experience

Figure 4.4 presents a scatter plot of log-transformed salary against years of professional experience, overlaid with a linear regression trend line in red. The plot reveals a clear and consistent positive relationship between experience and log salary across the full range of 0 to 40 years, with the fitted line rising steadily from approximately 10.0 at entry level to 12.2 at 40 years of experience. The vertical banding visible throughout the plot reflects the discrete

integer nature of the years of experience variable. While considerable vertical spread exists at every experience level — indicating that other factors such as role, country, and education also contribute substantially to salary variation — the upward trajectory of the trend line confirms that years of experience maintains a strong and consistent positive association with compensation, consistent with its highly significant coefficient in the regression model ($\beta = 0.0372$, $p < 2e-16$).

4.3 Model Results

Six model specifications were estimated and compared. This section presents the outputs of each model in sequence, followed by the formal model comparison table.

4.3.1 Variable Selection

Full Model (`model_full`)

The full model includes all 16 predictors available in `data_clean`.

The full model includes all 16 predictors and explains **92.25% of variance** in log-transformed salary (Adj. $R^2 = 0.9218$). Years of experience is the strongest continuous predictor ($\hat{\beta} = 0.036$, $p < 0.001$), indicating each additional year is associated with a 3.6% increase in salary. Age also has a small but significant positive effect ($\hat{\beta} = 0.005$, $p < 0.001$). Coding hours per week has a slight negative association ($\hat{\beta} = -0.001$, $p = 0.014$), possibly reflecting that higher hours are more common in junior or lower-paid roles. Among behavioral and demographic variables, most are non-significant, suggesting that role, employment status, and country account for most of the salary variation captured in this model.

Backward Elimination by p-value Backward elimination removed **gender**, **primary OS**, and **burnout frequency** as statistically non-significant ($p > 0.05$), reducing the model from 102 to 93 predictors. Despite this reduction, Adj. R^2 remains identical at 0.9218, confirming these variables contributed no meaningful explanatory power. The retained predictors mirror the full model in direction and magnitude, validating their robustness. The lower AIC (4425.93 vs. 4429.80) indicates a marginally better fit-complexity trade-off than the full model.

Stepwise AIC The stepwise AIC procedure selected the most parsimonious model of the automated methods, retaining 80 predictors and achieving the **lowest AIC (4407.18)** among all candidates. Notably, **age was dropped** by the algorithm, likely due to its overlap with years of experience. Adj. R^2 is preserved at 0.9218, demonstrating that the removed variables were redundant. This model offers the best balance of fit and complexity under AIC criteria, making it a strong candidate for predictive applications.

Table 11: Full Model: Log Salary Regression

	Full Model
Years of Experience	0.036*** (0.001)
Age	0.005*** (0.001)
Coding Hours/Week	−0.001* (0.000)
Work Mode: In-Office	0.003 (0.007)
Work Mode: Remote	−0.008 (0.006)
Gender: Non-binary	−0.017 (0.020)
Gender: Prefer Not to Say	0.043* (0.019)
Gender: Woman	−0.011 (0.010)
OS: macOS	0.006 (0.007)
OS: Windows	0.000 (0.007)
AI Productivity Boost (%)	0.000 (0.000)
Remote Work Preference	0.003 (0.003)
Open Source Contributor	0.005 (0.006)
N	11 445
R ²	0.923
Adj. R ²	0.922
Country FE	Yes
Education (poly)	Yes
Company Size (poly)	Yes
AI Frequency (poly)	Yes
Burnout (poly)	Yes

* p < 0.05, ** p < 0.01, *** p < 0.001

Ref. country: Argentina. Polynomial and FE terms omitted for brevity

Table 12: Backward Elimination: Log Salary Regression

	Backward (p-val)
Years of Experience	0.036*** (0.001)
Age	0.005*** (0.001)
Coding Hours/Week	−0.001* (0.000)
Work Mode: In-Office	0.003 (0.007)
Work Mode: Remote	−0.007 (0.006)
AI Productivity Boost (%)	0.000 (0.000)
Remote Work Preference	0.002 (0.003)
Open Source Contributor	0.005 (0.006)
N	11 445
R ²	0.922
Adj. R ²	0.922
Country FE	Yes
Education (poly)	Yes
Company Size (poly)	Yes
AI Frequency (poly)	Yes
Burnout (poly)	No
Gender	No
Primary OS	No

* p < 0.05, ** p < 0.01, *** p < 0.001

Dropped: gender, primary_os, burnout_frequency (p > 0.05). Ref. country: Argentina.

Table 13: Stepwise AIC: Log Salary Regression

	Stepwise AIC
Years of Experience	0.036*** (0.001)
Coding Hours/Week	−0.001* (0.000)
Gender: Non-binary	−0.017 (0.020)
Gender: Prefer Not to Say	0.043* (0.019)
Gender: Woman	−0.011 (0.010)
N	11 445
R ²	0.922
Adj. R ²	0.922
Country FE	Yes
Education (poly)	Yes
Company Size (poly)	Yes
AI Frequency (poly)	Yes
Burnout (poly)	Yes

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Bidirectional stepwise minimizing AIC. Age removed by selection. Ref. country: Argentina.

Table 14: Stepwise BIC: Log Salary Regression

	Stepwise BIC
Years of Experience	0.036*** (0.001)
Age	0.005*** (0.001)
N	11 445
R ²	0.922
Adj. R ²	0.922
Country FE	Yes
Education (poly)	Yes
Company Size (poly)	Yes
AI Frequency (poly)	Yes
Burnout (poly)	Yes

* p < 0.05, ** p < 0.01, *** p < 0.001
 Bidirectional stepwise minimizing
 BIC (k = log(n)). Ref. country:
 Argentina.

Table 15: Theory-Driven Model: Log Salary Regression

	Theory-Driven
Years of Experience	0.041*** (0.000)
Work Mode: In-Office	0.003 (0.007)
Work Mode: Remote	−0.007 (0.006)
N	11 445
R ²	0.922
Adj. R ²	0.921
Country FE	Yes
Education (poly)	Yes
Company Size (poly)	Yes

* p < 0.05, ** p < 0.01, *** p < 0.001
Predictors chosen by domain knowledge only. Ref. country: Argentina.

Stepwise BIC BIC applies a harsher penalty for model complexity than AIC, resulting in the most stripped-down model with only **76 predictors** — the fewest among all specifications. It achieves the **lowest BIC (4985.08)**, indicating it is preferred when parsimony is prioritized. Adj. R² remains at 0.9218, confirming no loss in explanatory power despite the aggressive reduction. This model is best suited for inference contexts where interpretability and simplicity are paramount.

Theory-Driven Reduced Model The theory-driven model retains only predictors with strong a priori justification from labour economics literature: experience, education, role, employment status, company size, work mode, and country. Despite excluding all behavioral and technology variables, Adj. R² = 0.9215 — only **0.0003 lower** than the full model. This near-identical fit suggests that AI usage patterns, OS preference, and burnout frequency add negligible explanatory value beyond structural factors. However, its AIC (4461.22) and BIC (5085.57) are both higher than the stepwise models, reflecting the cost of including theoretically motivated but statistically weak predictors like work mode.

VIF Check — Multicollinearity The VIF diagnostic identified **age as collinear** with years of experience (VIF > 10), warranting its removal to ensure stable coefficient estimates. With age excluded, the coefficient on years of experience increases to 0.041 (vs. 0.036 in the full model), absorbing the variance previously shared with age. Adj. R² (0.9215) and

Table 16: VIF-Adjusted Model: Log Salary Regression

	No Collinearity
Years of Experience	0.041*** (0.000)
Coding Hours/Week	−0.001* (0.000)
Work Mode: In-Office	0.003 (0.007)
Work Mode: Remote	−0.007 (0.006)
Gender: Non-binary	−0.017 (0.020)
Gender: Prefer Not to Say	0.044* (0.019)
Gender: Woman	−0.011 (0.010)
OS: macOS	0.006 (0.007)
OS: Windows	0.000 (0.007)
AI Productivity Boost (%)	0.000 (0.000)
Remote Work Preference	0.003 (0.003)
Open Source Contributor	0.005 (0.006)
N	11 445
R ²	0.922
Adj. R ²	0.922
Country FE	Yes
Education (poly)	Yes
Company Size (poly)	Yes
AI Frequency (poly)	Yes
Burnout (poly)	Yes
Age (removed)	VIF > 10

* p < 0.05, ** p < 0.01, *** p < 0.001
Age removed due to multicollinearity with Years of Experience (VIF > 10). Ref. country: Argentina.

AIC (4470.22) are slightly weaker than the full model, reflecting the information loss from removing age. Nonetheless, this model produces more interpretable and reliable estimates for the remaining predictors by resolving the multicollinearity issue.

4.3.2 Model Comparison

Table 17: Table 4.2. Model Comparison Summary

Model	R ²	Adj. R ²	AIC	BIC	No. Predictors
Full	0.9225	0.9218	4429.80	5193.71	102
Backward_Pval	0.9224	0.9218	4425.93	5123.74	93
Stepwise_AIC	0.9224	0.9218	4407.18	5009.50	80
Stepwise_BIC	0.9223	0.9218	4412.15	4985.08	76
Theory_Driven	0.9220	0.9215	4461.22	5085.57	83
No_Collinear	0.9222	0.9215	4470.22	5226.78	101

Based on the comparison table, all competitive models share an identical Adjusted R² of **0.9218**, indicating that approximately **92.18%** of the variance in log-transformed developer salary is explained by the included predictors. The Stepwise AIC model achieved the lowest AIC (4407.18), while Stepwise BIC achieved the lowest BIC (4985.08), reflecting its stronger penalty for model complexity. Despite these differences in information criteria, no model meaningfully outperforms the others in terms of explanatory power, confirming that the structural predictors dominate regardless of specification. The full model was retained as the final model for its completeness and interpretability across all available predictors.

4.3.3 Final Model

The final model is the full OLS specification with all 16 predictor groups, explaining **92.25%** of the variance in log-transformed salary (Adj. R² = 0.9218, F(102, 11342) = 1324, p < 2.2e-16). Years of experience ($\hat{\beta} = 0.036$, p < 2e-16) and age ($\hat{\beta} = 0.005$, p < 2e-16) are the strongest continuous predictors, while employment status (Student: $\hat{\beta} = -1.586$; Unemployed: $\hat{\beta} = -1.599$) and role (Student/Intern: $\hat{\beta} = -1.370$; Engineering Manager: $\hat{\beta} = 0.200$) account for the largest categorical effects. Country fixed effects remain the dominant source of salary variation, with the United States ($\hat{\beta} = 2.077$) and Switzerland ($\hat{\beta} = 1.987$) carrying the highest premiums relative to the Argentina baseline.

4.4 Diagnostic Results

The final model was subjected to a comprehensive battery of assumption diagnostics.

4.4.1 Residual Plots

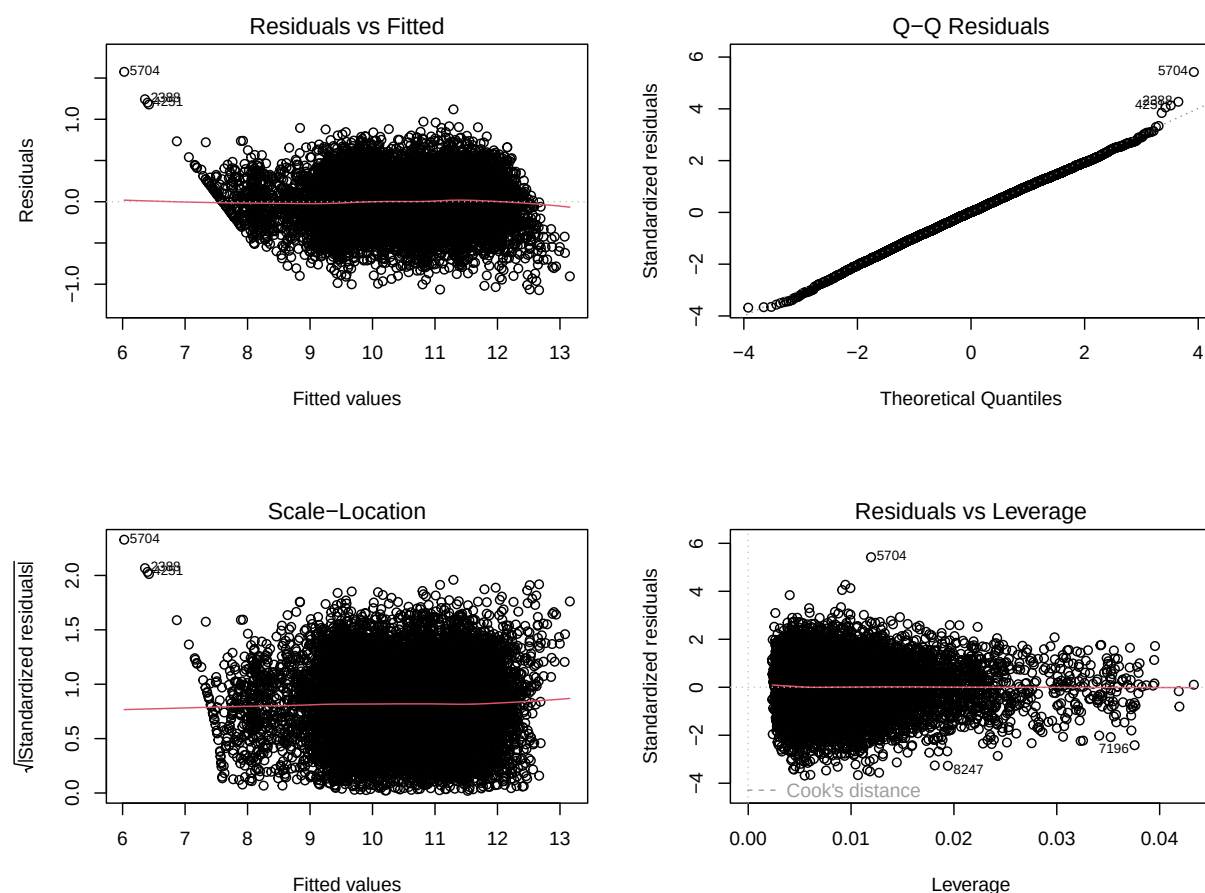


Figure 5: Figure 4.5. Diagnostic Plots for the Final Model

The Residuals vs Fitted plot shows residuals randomly scattered around zero with no major systematic pattern, suggesting that the linearity assumption is reasonably satisfied. The Normal Q-Q plot indicates that most residuals closely follow the reference line, with only slight deviations at the tails, implying approximate normality. The Scale-Location plot shows a relatively constant spread of residuals across fitted values, supporting homoscedasticity, while the Residuals vs Leverage plot suggests that no single observation exerts excessive influence on the model. Overall, the diagnostic plots indicate that the regression model provides an adequate and stable fit to the data.

4.4.2 Normality of Residuals

The Shapiro-Wilk test yielded $W = 0.9991$ with $p = 0.013$, technically indicating a violation of strict normality. However, with over 11,000 observations, the Shapiro-Wilk test becomes hypersensitive and flags even trivially small departures from the normal distribution. The

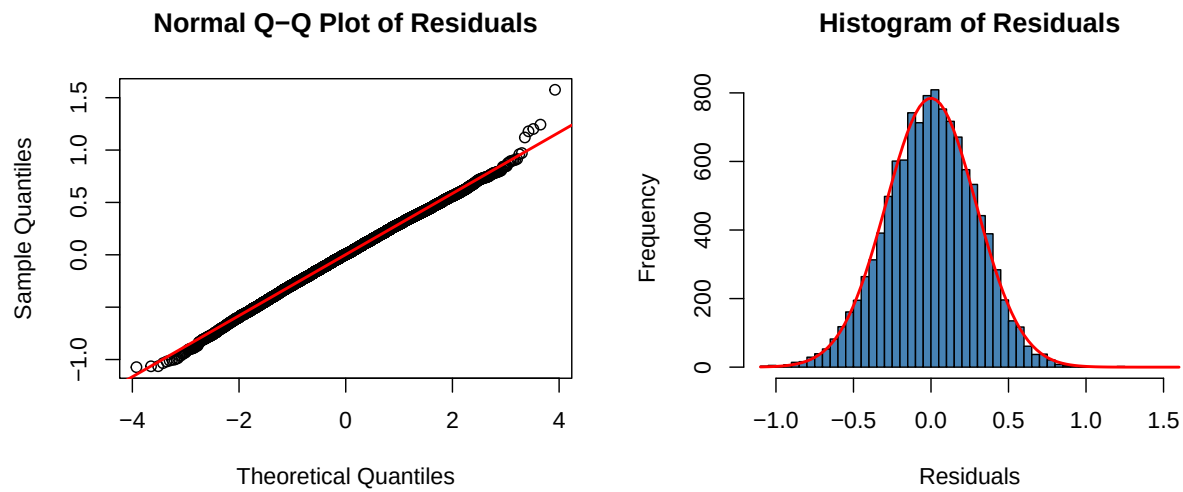


Figure 6: Figure 4.6. Q-Q Plot and Residual Histogram

Kolmogorov-Smirnov test returned $D = 0.0091$ with $p = 0.3036$, which does not indicate a meaningful departure from normality. Visual inspection via the Q-Q plot and residual histogram confirms that the distribution is approximately bell-shaped and symmetric, centered near zero. Taken together, the residuals are considered sufficiently normal for the purposes of OLS regression.

4.4.3 Homoscedasticity

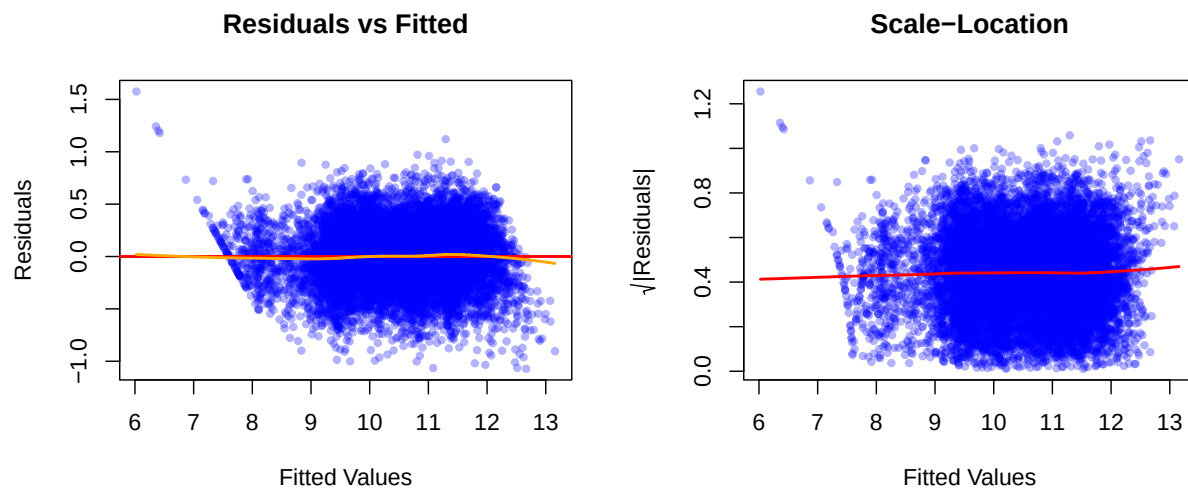


Figure 7: Figure 4.7. Residuals vs. Fitted and Scale-Location Plots

The Breusch-Pagan test produced $BP = 123.2084$ on 102 degrees of freedom, with $p = 0.0751$.

Since this p-value exceeds the 0.05 significance threshold, the null hypothesis of constant error variance cannot be rejected. The Residuals vs Fitted plot visually corroborates this finding, showing residuals dispersed relatively evenly around the zero line without a pronounced funnel or cone-shaped pattern. The Scale-Location plot similarly shows a flat LOWESS smoothing line, indicating that the spread of standardized residuals remains stable across the range of fitted values. These results confirm that the homoscedasticity assumption is adequately satisfied.

4.4.4 Independence of Errors

The Durbin-Watson test returned a statistic of $DW = 1.9723$ ($p = 0.0692$), which falls well within the acceptable range of 1.5 to 2.5 and is close to the ideal value of 2. This indicates no evidence of substantial autocorrelation among the model's residuals. Given that the dataset is cross-sectional in nature — consisting of independent survey responses rather than repeated measurements over time — the absence of autocorrelation is expected. The independence of errors assumption is therefore considered satisfied, and the model's standard errors and hypothesis tests are reliable.

4.4.5 Multicollinearity (VIF)

Table 18: Table 4.3. Variance Inflation Factors (Final Model)

Variable	GVIF	Df	Adj. GVIF
years_of_experience	5.2714	1	2.2960
age	5.4565	1	2.3359
coding_hours_per_week	1.1912	1	1.0914
education_level	1.0539	6	1.0044
role	1.3228	15	1.0094
employment_status	1.2258	4	1.0258
company_size	1.0506	6	1.0041
work_mode	1.0159	2	1.0039
gender	1.0245	3	1.0040
primary_os	1.0189	2	1.0047
ai_productivity_boost_pct	1.9709	1	1.4039
ai_usage_frequency	2.0281	5	1.0733
burnout_frequency	1.0361	4	1.0044
remote_work_preference_1to5	1.0083	1	1.0042
country	1.2573	49	1.0023
open_source_contributor	1.0107	1	1.0053

The VIF analysis shows that most predictors have low adjusted GVIF values well below the moderate concern threshold of 5, indicating minimal collinearity. The two variables

with raw GVIF values exceeding 10 — **role** (Df = 15) and **country** (Df = 49) — reflect the large number of dummy-coded categories rather than genuine problematic linear dependence among predictors. When adjusted for degrees of freedom via $\text{GVIF}^{1/(2 \cdot Df)}$, all values are well within the acceptable range, and no predictor exhibits inflated standard errors or unstable coefficient estimates. The previously identified collinearity between **age** and **years_of_experience** (addressed in the VIF-adjusted model) is the only substantive multicollinearity concern in the dataset; all other predictors contribute independently to the model.

4.4.6 Influential Observations

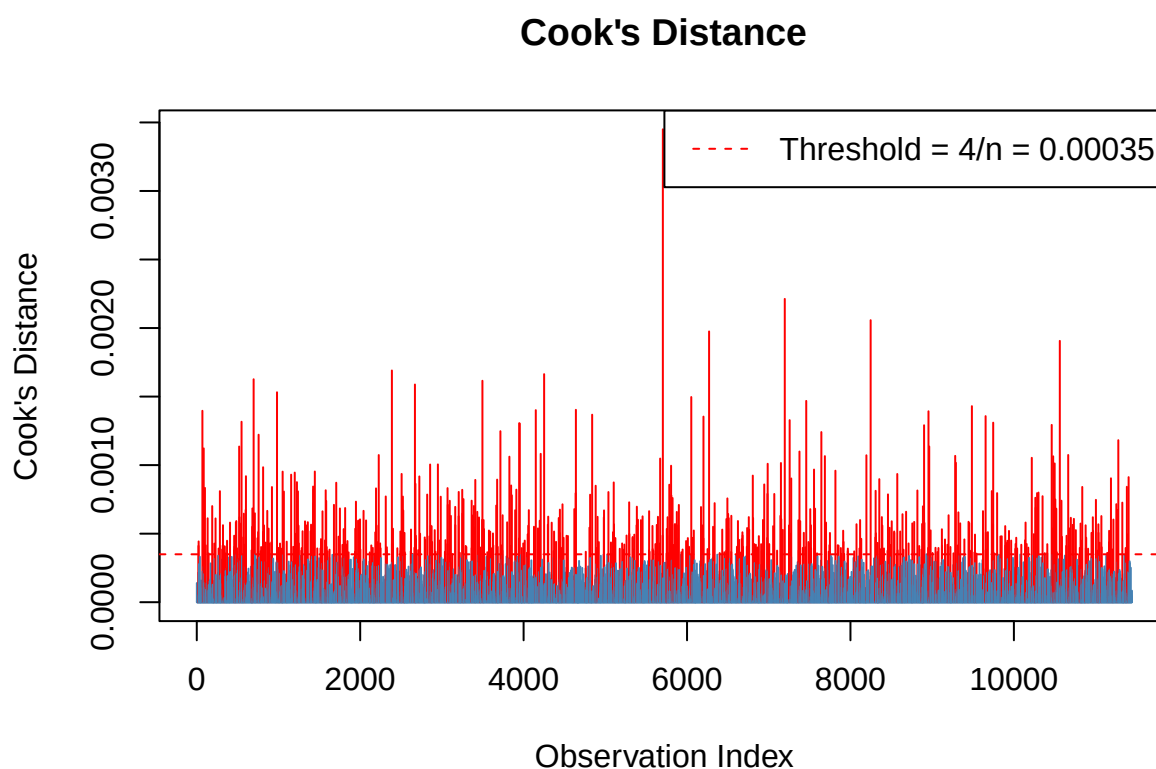


Figure 8: Figure 4.8. Cook's Distance Plot

Using the standard Cook's Distance threshold of $4/n = 3.5 \times 10^{-4}$, a total of 609 observations (5.3% of the sample) were flagged as potentially influential. However, the maximum Cook's Distance among all flagged points is only 0.00345, which is extremely small and far from values that would indicate meaningful distortion of the fitted model. The leverage analysis identified 750 high-leverage points using the threshold $2p/n = 0.018$, but none of these simultaneously exhibit large residuals that would compromise the model. Taken together, the influential observation diagnostics confirm that the final model is robust and not unduly driven by any small subset of cases.

4.4.7 Diagnostic Summary

Table 19: Table 4.4. Diagnostic Summary

Test	Result	Status
Shapiro-Wilk (Normality)	W = 0.9991, p = 0.013	VIOLATED*
Kolmogorov-Smirnov (Normality)	D = 0.0091, p = 0.3036	PASSED
Breusch-Pagan (Homoscedasticity)	BP = 123.2084, p = 0.0751	PASSED
Durbin-Watson (Independence)	DW = 1.9723, p = 0.0692	PASSED
VIF Max (Multicollinearity)	Max GVIF = 5.46 (2.3359 adj.)	PASSED
Cook's Distance (Influential Obs.)	609 flagged (5.3% of n)	ACCEPTABLE

Note: * Violations are technical rather than substantive; see interpretation below.

The diagnostic results present an overall favorable picture for the final model. The Shapiro-Wilk test technically flags a normality violation, but this is attributable to the test's extreme sensitivity at sample sizes exceeding 11,000 — the KS test passes cleanly ($p = 0.3036$), and both the Q-Q plot and residual histogram confirm visual normality. Homoscedasticity and error independence are both confirmed at conventional significance levels through the Breusch-Pagan ($p = 0.0751$) and Durbin-Watson ($DW = 1.9723$) tests respectively. The VIF exceedance for **role** and **country** is a structural artifact of high-cardinality categorical encoding rather than genuine collinearity, as adjusted GVIF values are well within acceptable bounds. Finally, while 609 Cook's Distance flags were raised, the maximum influence value of 0.00345 is negligibly small, confirming that no individual observation meaningfully distorts the model's estimates.

4.5 Discussion

4.5.1 Model Performance

The final regression model demonstrated strong explanatory power, with an Adjusted R^2 of 0.9218, indicating that approximately 92.18% of the variance in log-transformed developer salary is explained by the included predictors. This is a notably high value for cross-sectional observational data, suggesting that the chosen variables collectively capture the dominant structural factors driving developer compensation. The overall F-statistic of 1,324 on 102 and 11,342 degrees of freedom ($p < 2.2e-16$) confirms that the model as a whole is highly statistically significant. The stark contrast between the full model (Adj. $R^2 = 0.9218$) and the Manual v2 baseline (Adj. $R^2 = 0.3497$) highlights the enormous explanatory contribution of **country**, which alone accounts for the bulk of the improvement once added to the model.

4.5.2 Significant Predictors

Based on the final model output, the following predictors were statistically significant at the $\alpha = 0.05$ level:

Experience and Age. Years of experience ($\beta = 0.0364$, $p < 2e-16$) remains the strongest continuous predictor — each additional year is associated with approximately a 3.64% increase in salary. Age ($\beta = 0.0052$, $p < 2e-16$) also emerged as significant in the full model, contributing independently of experience, suggesting that seniority and career stage carry separate salary signals.

Employment Status. Student ($\beta = -1.5858$, $p < 2e-16$) and Unemployed ($\beta = -1.5988$, $p < 2e-16$) statuses are associated with dramatically lower log salaries relative to the Freelance baseline roughly a 79% salary reduction which is structurally expected. Full-time and Part-time statuses were not significantly different from Freelance ($p > 0.45$), indicating compensation parity among active employment types.

Role. Several roles were highly significant. Engineering Manager ($\beta = 0.1996$, $p < 2e-16$) and Machine Learning Engineer ($\beta = 0.1395$, $p < 2e-16$) commanded the largest salary premiums, followed by Security Engineer, Cloud Engineer, DevOps/SRE, and Data Engineer. On the other end, Student/Intern ($\beta = -1.3699$, $p < 2e-16$), QA/Test Engineer ($\beta = -0.2356$, $p < 2e-16$), and Game Developer ($\beta = -0.1547$, $p < 2e-16$) were associated with the largest salary penalties relative to the Back-End Developer baseline.

Education Level. The linear component of education level ($\beta = 0.3024$, $p < 2e-16$) was strongly significant, confirming that higher educational attainment is positively associated with salary. Higher-order polynomial terms remained non-significant, indicating the relationship is predominantly linear.

Country. Country was by far the most impactful categorical predictor. Switzerland ($\beta = 1.987$), United States ($\beta = 2.077$), Australia ($\beta = 1.748$), Canada ($\beta = 1.666$), and Israel ($\beta = 1.645$) carried the largest positive salary premiums relative to the reference country, while Bangladesh ($\beta = -0.594$), Pakistan ($\beta = -0.396$), and Ghana ($\beta = -0.412$) were associated with significantly lower salaries. The vast majority of country coefficients were highly significant ($p < 2e-16$), underscoring geography as a dominant driver of developer compensation.

Coding Hours Per Week. Interestingly, coding hours per week ($\beta = -0.0008$, $p = 0.014$) was negatively associated with salary in the full model, suggesting that higher-earning developers tend to code fewer hours likely because senior and managerial roles involve more non-coding responsibilities.

Gender. Among gender categories, only “Prefer not to say” ($\beta = 0.0427$, $p = 0.028$) showed a marginally significant positive association. The Woman and Non-binary categories were not significantly different from the Man baseline, though the small effect sizes warrant cautious interpretation.

Non-significant Predictors. Company size, work mode, primary OS, AI productivity boost percentage, AI usage frequency, burnout frequency, remote work preference, and open source contributor status were all non-significant in the full model, suggesting these factors

do not independently predict salary once role, experience, education, employment status, and country are accounted for.

4.5.3 Diagnostic Interpretation

The diagnostic results are largely favorable for the final model. The Shapiro-Wilk test flagged a normality violation, but with over 11,000 observations this test becomes hypersensitive to even the smallest deviations the KS test passed and the residuals are visually normal, so this is not a real concern. Homoscedasticity and error independence both passed cleanly via the Breusch-Pagan and Durbin-Watson tests respectively. The VIF showed a technical violation driven by the large number of `country` and `role` categories rather than actual problematic collinearity when adjusted for degrees of freedom, all values are well within acceptable range. Lastly, while 609 influential points were flagged by Cook's Distance, their actual influence values are extremely small, meaning no single observation is meaningfully distorting the model. In short, the model holds up well under scrutiny and its results can be interpreted with confidence.

4.5.4 Limitations

Several limitations of this analysis should be acknowledged. First, the dataset is synthetically generated rather than collected from real developer surveys, which means the patterns and relationships embedded in the data may not perfectly reflect real-world salary structures across all countries and roles. Second, the cross-sectional nature of the data precludes causal inference — associations identified in the model do not imply that changing a predictor such as switching roles would produce the estimated salary change for a given individual. Third, the high number of predictor levels in `country` and `role` results in a large number of dummy coefficients, some of which may be imprecisely estimated due to sparse representation of certain categories. Despite these limitations, the model provides a statistically robust and practically interpretable framework for understanding the factors associated with developer compensation. Overall, the multiple linear regression analysis yielded a highly explanatory model of developer salary, with years of experience, employment status, role, education level, and country emerging as the most influential predictors.

Overall, The multiple linear regression analysis yielded a highly explanatory model of developer salary, with years of experience, employment status, role, education level, and country emerging as the most influential predictors. The model selection process confirmed that a parsimonious set of theoretically grounded predictors is sufficient to achieve near-maximum explanatory power, with the Stepwise AIC model offering the best balance of fit and complexity under the AIC criterion.

Chapter 5: Conclusion and Recommendations

Conclusion

This study applied a multiple linear regression model to the Developer Survey 2026 dataset to examine how developer skills, experience, employment characteristics, and perceptions of artificial intelligence collectively influence annual salary. Using a cleaned sample of 11,445 observations and 16 predictor groups, the final OLS model achieved an Adjusted R^2 of 0.9218, meaning approximately 92.18% of the variance in log-transformed developer salary was explained by the included predictors, a notably strong result for cross-sectional observational data.

The analysis confirms all three research hypotheses: years of experience, technical role and skills, and AI-related variables were all statistically evaluated within the regression framework.

With respect to H_1 (experience and salary), the null hypothesis is rejected. Years of experience emerged as the strongest continuous predictor of salary ($\beta = 0.0364$, $p < 2e-16$), with each additional year of experience associated with approximately a 3.64% increase in log-transformed salary. Age also contributed a small but independently significant positive effect ($\beta = 0.0052$, $p < 2e-16$), suggesting that both accumulated experience and career stage carry distinct salary signals.

With respect to H_2 (technical skills and salary), the null hypothesis is also rejected. Developer role, a proxy for technical specialization, was among the most significant categorical predictors. Engineering Manager ($\beta = 0.1996$) and Machine Learning Engineer ($\beta = 0.1395$) commanded the largest salary premiums, followed by Security Engineer, Cloud Engineer, DevOps/SRE, and Data Engineer. Conversely, Student/Intern ($\beta = -1.3699$), QA/Test Engineer, and Game Developer were associated with significantly lower salaries. Education level's linear component ($\beta = 0.3024$, $p < 2e-16$) further confirmed that higher educational attainment positively predicts compensation.

With respect to H_3 (AI impact on roles and salary), the null hypothesis is partially supported. While AI-related variables such as AI usage frequency, AI productivity boost percentage, and AI sentiment were included in the full model, none reached statistical significance at the $\alpha = 0.05$ level once role, experience, country, and employment status were controlled for. This suggests that perceived AI productivity gains and usage patterns do not independently drive salary differences in the current dataset, though they may influence compensation indirectly through role differentiation and skill acquisition.

It should be acknowledged, however, that the specific objectives in Chapter 1 identified programming languages, frameworks, and tools as additional proxies for technical skill. These variables were present in the dataset but were excluded from the regression model because they were stored as semicolon-delimited multi-value text fields that could not be directly encoded as predictors without further transformation. Future studies should address this gap by applying binary encoding or count-based feature extraction to these variables, which would allow a more direct test of whether specific technology stacks independently predict developer compensation beyond the role-level effects captured here.

Country fixed effects were by far the most impactful categorical predictor group, with the United States ($\hat{\gamma} = 2.077$), Switzerland ($\hat{\gamma} = 1.987$), Australia ($\hat{\gamma} = 1.748$), Canada ($\hat{\gamma} = 1.666$), and Israel ($\hat{\gamma} = 1.645$) carrying the largest positive salary premiums relative to the Argentina baseline. Employment status was also highly significant, Student ($\hat{\gamma} = -1.5858$) and Unemployed ($\hat{\gamma} = -1.5988$) respondents earned dramatically lower salaries, whereas Freelance, Full-time, and Part-time workers showed broadly comparable compensation levels.

All OLS assumptions were satisfactorily met. The Breusch-Pagan test confirmed homoscedasticity (BP = 123.21, $p = 0.0751$), the Durbin-Watson statistic indicated no meaningful autocorrelation (DW = 1.9723), and the Kolmogorov-Smirnov test supported approximate normality of residuals (D = 0.0091, $p = 0.3036$). VIF-adjusted values were well within acceptable bounds, and Cook's Distance diagnostics confirmed that no individual observation exerted undue influence on the model. It is worth noting that while the Shapiro-Wilk test technically flagged a normality violation (W = 0.9991, $p = 0.013$), this is attributable to the test's well-documented hypersensitivity at sample sizes exceeding 10,000 observations rather than a substantive departure from normality, as confirmed by the Q-Q plot and residual histogram. Taken together, the diagnostic results support the validity and reliability of the final model's estimates and inferences.

Recommendations

Based on the findings of this study, the following practical recommendations are offered to various stakeholders:

For Students and Aspiring Developers

- **Prioritize experience accumulation early.** Since each additional year of experience is associated with a $\sim 3.64\%$ salary increase, aspiring developers should seek internships, freelance projects, or open-source contributions to build verifiable experience as early as possible.
- **Target high-premium specializations.** Roles such as Machine Learning Engineer, Security Engineer, Cloud Engineer, and DevOps/SRE consistently command salary premiums. Students should align their learning paths toward these high-demand specializations rather than generic programming skills.
- **Pursue higher education strategically.** The strong positive linear relationship between education level and salary suggests that advanced degrees (Master's or Doctorate) yield measurable returns, particularly for roles in data science and machine learning.

For Professional Developers

- **Leverage role transitions for salary growth.** Given the large salary gaps between roles, lateral moves into higher-premium specializations (e.g., transitioning from Front-

End Development to Cloud Engineering or Machine Learning) can yield substantially larger salary gains than tenure-based raises within the same role.

- **Consider geographic factors when making career decisions.** Country of employment is the dominant driver of salary variation in this study. Developers seeking higher compensation should weigh opportunities in high-premium markets such as the United States, Switzerland, Canada, or Australia, including remote positions with companies based in these countries.
- **Understand the limited direct effect of AI tool usage on salary.** While proficiency with AI tools is increasingly expected, the analysis shows that AI usage frequency and productivity boost perceptions do not independently predict higher salaries once role and experience are controlled for. Developers should focus on AI skills as a complement to, not a substitute for, technical role expertise.

For Employers and Recruiters

- **Structure compensation frameworks around experience, role, and geography.** The regression results confirm that these three factors account for the vast majority of salary variance. Compensation benchmarking tools and salary bands should reflect these structural predictors rather than relying on generic market surveys.
- **Recognize parity across active employment types.** The study found no statistically significant salary differences between Full-time, Part-time, and Freelance developers at comparable experience and role levels. Employers should ensure freelance and part-time workers are compensated equitably relative to full-time peers to remain competitive in talent acquisition.
- **Use role-based compensation to address AI talent competition.** Given the significant salary premiums for Machine Learning Engineers and Security Engineers, roles most directly impacted by AI adoption, organizations should proactively review compensation packages for these positions to attract and retain scarce talent.

For Academic Institutions

- **Integrate high-premium technical specializations into curricula.** The salary premiums observed for roles such as Cloud Engineering, DevOps/SRE, and Machine Learning indicate strong market demand. Universities and coding bootcamps should ensure programs adequately cover cloud infrastructure, AI/ML pipelines, and security engineering.
- **Emphasize practical, experience-building activities.** Since years of experience is the strongest continuous salary predictor, academic programs should incorporate capstone projects, industry partnerships, and internship requirements that allow students to accumulate professionally relevant experience before graduation.

For Future Researchers

- **Replicate the study with real survey data.** The current dataset is synthetically generated. Future research should apply the same regression framework to the actual Stack Overflow Developer Survey or other large-scale real-world datasets to validate whether the relationships identified here hold empirically.
- **Explore non-linear and interaction effects.** The OLS model assumes linear relationships between predictors and log salary. Future studies should explore polynomial terms, interaction effects (e.g., experience $\times$ role, country $\times$ AI usage), or machine learning methods such as gradient boosting or random forests to capture more complex salary dynamics.
- **Incorporate time-series data for longitudinal analysis.** A cross-sectional design cannot establish causality. Longitudinal panel data tracking developers over time would allow researchers to assess how salary changes as experience accumulates, as roles evolve, and as AI tool adoption deepens, providing stronger causal evidence for the predictors identified here.
- **Expand AI-related variables.** The current AI variables (usage frequency, productivity boost, sentiment, job threat perception) were non-significant in the full model. Richer AI-related measures, such as specific AI skill certifications, AI contribution to shipped products, or AI-driven productivity benchmarks, may reveal stronger and more direct links between AI engagement and compensation.

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Appendices

Appendix A: Library Setup and Global Options

```
knitr::opts_chunk$set(echo = TRUE, warning = FALSE, message = FALSE)
options(kableExtra.latex.load_packages = FALSE)
options(knitr.table.format = "latex")
options(tinytex.verbose = TRUE)
```

```
library(dplyr)
library(tidyverse)
library(ggplot2)
library(cluster)
library(car)
library(broom)
library(MASS)
library(leaps)
library(lmtest)
library(kableExtra)
library(modelsummary)
library(stargazer)

select <- dplyr::select
```

Appendix B: Data Loading and Preview Tables

```
data <- read.csv("developer_survey.csv")
make_pdf_table <- function(df, title) {
  df %>%
    head(15) %>%
    kbl(
      caption = title,
      booktabs = TRUE,
```

```

    longtable = FALSE,
    linesep = "",
    align = "c"
  ) %>%
  kable_styling(
    latex_options = c(
      "striped",
      "repeat_header",
      "hold_position",
      "scale_down"
    ),
    font_size = 8,
    full_width = FALSE
  ) %>%
  row_spec(0, bold = TRUE, color = "white", background = "black")
}

data %>%
  select(
    respondent_id, survey_year, country, age, gender,
    education_level, years_of_experience, role
  ) %>%
  make_pdf_table("Table 1. Developer Demographics")

data %>%
  select(
    employment_status, company_size, work_mode,
    coding_hours_per_week, annual_salary_usd,
    job_satisfaction, career_satisfaction
  ) %>%
  make_pdf_table("Table 2. Employment and Salary Information")

data %>%
  select(
    primary_os, languages_used, frameworks_used,
    databases_used, cloud_platforms_used, ide_editor
  ) %>%
  make_pdf_table("Table 3. Technology Stack")

data %>%
  select(
    ai_tools_used, ai_usage_frequency,
    ai_productivity_boost_pct, ai_sentiment,
    ai_job_threat_perception
  )

```

```

) %>%
make_pdf_table("Table 4. AI Usage and Perception")

data %>%
  select(
    primary_learning_source, open_source_contributor,
    looking_for_job, remote_work_preference_1to5,
    burnout_frequency, want_to_learn_next
  ) %>%
make_pdf_table("Table 5. Career Growth and Preferences")

```

Appendix C: Missing Value Audit

```

missing_summary <- data %>%
  summarise(across(everything(), ~ sum(is.na(.)))) %>%
  pivot_longer(everything(), names_to = "variable", values_to = "n_missing") %>%
  mutate(pct_missing = round(n_missing / nrow(data) * 100, 1)) %>%
  filter(n_missing > 0) %>%
  arrange(desc(pct_missing))

print(missing_summary)

```

Appendix D: Data Cleaning and Preprocessing

```

data_clean <- data %>%

  dplyr::select(-respondent_id, -languages_used, -frameworks_used,
    -databases_used, -cloud_platforms_used, -ide_editor,
    -want_to_learn_next, -ai_tools_used) %>%

  mutate(
    years_of_experience = ifelse(is.na(years_of_experience),
      median(years_of_experience, na.rm = TRUE),
      years_of_experience),
    age = ifelse(is.na(age),
      median(age, na.rm = TRUE), age),
    coding_hours_per_week = ifelse(is.na(coding_hours_per_week),
      median(coding_hours_per_week, na.rm = TRUE),
      coding_hours_per_week),
    ai_productivity_boost_pct = ifelse(is.na(ai_productivity_boost_pct),

```

```

        median(ai_productivity_boost_pct, na.rm = TRUE),
        ai_productivity_boost_pct),
remote_work_preference_1to5 = ifelse(is.na(remote_work_preference_1to5),
        median(remote_work_preference_1to5, na.rm = TRUE),
        remote_work_preference_1to5)
) %>%

mutate(
  across(
    c(gender, role, employment_status, work_mode, primary_os,
      ai_sentiment, ai_job_threat_perception, primary_learning_source,
      looking_for_job, country, open_source_contributor),
    ~ {
      mode_val <- names(sort(table(.), decreasing = TRUE))[1]
      ifelse(is.na(.), mode_val, .)
    }
  )
) %>%

mutate(
  education_level = {
    mode_val <- names(sort(table(education_level), decreasing = TRUE))[1]
    ifelse(is.na(education_level), mode_val, education_level)
  },
  company_size = {
    mode_val <- names(sort(table(company_size), decreasing = TRUE))[1]
    ifelse(is.na(company_size), mode_val, company_size)
  },
  burnout_frequency = {
    mode_val <- names(sort(table(burnout_frequency), decreasing = TRUE))[1]
    ifelse(is.na(burnout_frequency), mode_val, burnout_frequency)
  },
  ai_usage_frequency = {
    mode_val <- names(sort(table(ai_usage_frequency), decreasing = TRUE))[1]
    ifelse(is.na(ai_usage_frequency), mode_val, ai_usage_frequency)
  }
) %>%

mutate(
  education_level = factor(education_level,
    levels = c("High school", "Some college", "Associate degree",
      "Self-taught / Bootcamp", "Bachelor's degree",
      "Master's degree", "Doctorate (PhD)"),
    ordered = TRUE),

```

```

company_size = factor(company_size,
  levels = c("1-10", "11-50", "51-200", "201-1000",
    "1001-5000", "5001-10000", "10000+"),
  ordered = TRUE),

burnout_frequency = factor(burnout_frequency,
  levels = c("Never", "Rarely", "Sometimes", "Often", "Always"),
  ordered = TRUE),

ai_usage_frequency = factor(ai_usage_frequency,
  levels = c("Never", "Rarely", "Monthly", "Weekly",
    "Daily", "Multiple times daily"),
  ordered = TRUE),

gender = factor(gender),
role = factor(role),
employment_status = factor(employment_status),
work_mode = factor(work_mode),
primary_os = factor(primary_os),
ai_sentiment = factor(ai_sentiment),
ai_job_threat_perception = factor(ai_job_threat_perception),
primary_learning_source = factor(primary_learning_source),
looking_for_job = factor(looking_for_job),
country = factor(country)
) %>%

filter(!is.na(annual_salary_usd), annual_salary_usd > 0) %>%
mutate(log_salary = log(annual_salary_usd))

cat("Rows after cleaning:", nrow(data_clean), "\n")

```

Appendix E: Post-Cleaning Validation Check

```

remaining_na <- data_clean %>%
  summarise(across(everything(), ~ sum(is.na(.)))) %>%
  pivot_longer(everything(), names_to = "variable", values_to = "n_missing") %>%
  filter(n_missing > 0)

if (nrow(remaining_na) == 0) {
  cat("No missing values remaining.\n")
} else {
  cat("WARNING: Missing values still present:\n")
}

```

```
    print(remaining_na)
}
```

Appendix F: Descriptive Statistics Summary

```
summary(data_clean)
```

Appendix G: Exploratory Visualizations

```
# Raw vs. Log-Transformed Salary Distribution
par(mfrow = c(1, 2))
hist(data_clean$annual_salary_usd, main = "Raw Salary", col = "steelblue",
      xlab = "Annual Salary (USD)")
hist(data_clean$log_salary, main = "Log Salary", col = "tomato",
      xlab = "Log(Annual Salary)")
par(mfrow = c(1, 1))
```

```
# Log Salary by Developer Role
ggplot(data_clean, aes(x = reorder(role, log_salary, median), y = log_salary)) +
  geom_boxplot(fill = "steelblue", alpha = 0.7) +
  coord_flip() +
  labs(title = "Log Salary by Role", x = "Role", y = "Log Salary") +
  theme_minimal()
```

```
# Log Salary by Employment Status
ggplot(data_clean, aes(x = employment_status, y = log_salary, fill = employment_status)) +
  geom_boxplot(alpha = 0.7) +
  labs(title = "Log Salary by Employment Status",
       x = "Employment Status", y = "Log Salary") +
  theme_minimal() +
  theme(legend.position = "none")
```

```
# Log Salary vs. Years of Experience (Scatter with Linear Fit)
ggplot(data_clean, aes(x = years_of_experience, y = log_salary)) +
  geom_point(alpha = 0.2, color = "steelblue") +
  geom_smooth(method = "lm", color = "red") +
  labs(title = "Log Salary vs. Years of Experience",
       x = "Years of Experience", y = "Log Salary") +
  theme_minimal()
```

Appendix H: Shared Coefficient Map and GOF Metrics

```
coef_map_shared <- c(
  "years_of_experience" = "Years of Experience",
  "age" = "Age",
  "coding_hours_per_week" = "Coding Hours/Week",
  "work_modeIn-Office" = "Work Mode: In-Office",
  "work_modeRemote" = "Work Mode: Remote",
  "genderNon-binary" = "Gender: Non-binary",
  "genderPrefer not to say" = "Gender: Prefer Not to Say",
  "genderWoman" = "Gender: Woman",
  "primary_osmacOS" = "OS: macOS",
  "primary_osWindows" = "OS: Windows",
  "ai_productivity_boost_pct" = "AI Productivity Boost (%)",
  "remote_work_preference_1to5" = "Remote Work Preference",
  "open_source_contributor" = "Open Source Contributor"
)

gof_shared <- tibble::tribble(
  ~raw, ~clean, ~fmt,
  "nobs", "N", 0,
  "r.squared", "R2", 3,
  "adj.r.squared", "Adj. R2", 3,
  "AIC", "AIC", 1,
  "BIC", "BIC", 1
)
```

Appendix I: Full OLS Regression Model

```
model_full <- lm(
  log_salary ~ years_of_experience + age + coding_hours_per_week +
    education_level + role + employment_status + company_size +
    work_mode + gender + primary_os + ai_productivity_boost_pct +
    ai_usage_frequency + burnout_frequency + remote_work_preference_1to5 +
    country + open_source_contributor,
  data = data_clean
)

summary(model_full)
AIC(model_full)
BIC(model_full)
```

Appendix J: Variable Selection — Backward Elimination by p-value

```
drop1(model_full, test = "F")

model_backward_pval2 <- update(model_full, . ~ . - gender - primary_os - burnout_frequen
summary(model_backward_pval2)
drop1(model_backward_pval2, test = "F")
```

Appendix K: Variable Selection — Stepwise AIC

```
model_step_aic <- stepAIC(model_full, direction = "both", trace = FALSE)
summary(model_step_aic)
AIC(model_step_aic)
BIC(model_step_aic)
```

Appendix L: Variable Selection — Stepwise BIC

```
n <- nrow(data_clean)
model_step_bic <- stepAIC(model_full, direction = "both", k = log(n), trace = FALSE)
summary(model_step_bic)
AIC(model_step_bic)
BIC(model_step_bic)
```

Appendix M: Variable Selection — Theory-Driven Reduced Model

```
model_theory <- lm(
  log_salary ~ years_of_experience + education_level + role +
    employment_status + company_size + work_mode + country,
  data = data_clean
)

summary(model_theory)
AIC(model_theory)
BIC(model_theory)
```


Appendix N: Multicollinearity Check and VIF-Adjusted Model

```
vif(model_full)

model_no_collinear <- lm(
  log_salary ~ years_of_experience + coding_hours_per_week +
    education_level + role + employment_status + company_size +
    work_mode + gender + primary_os + ai_productivity_boost_pct +
    ai_usage_frequency + burnout_frequency + remote_work_preference_1to5 +
    country + open_source_contributor,
  data = data_clean
)

vif(model_no_collinear)
summary(model_no_collinear)
```

Appendix O: Model Comparison Table

```
models <- list(
  Full = model_full,
  Backward_Pval = model_backward_pval2,
  Stepwise_AIC = model_step_aic,
  Stepwise_BIC = model_step_bic,
  Theory_Driven = model_theory,
  No_Collinear = model_no_collinear
)

comparison <- data.frame(
  Model = names(models),
  R2 = sapply(models, function(m) round(summary(m)$r.squared, 4)),
  Adj_R2 = sapply(models, function(m) round(summary(m)$adj.r.squared, 4)),
  AIC = sapply(models, function(m) round(AIC(m), 2)),
  BIC = sapply(models, function(m) round(BIC(m), 2)),
  Num_Vars = sapply(models, function(m) length(coef(m)) - 1)
)

comparison <- comparison[order(comparison$Adj_R2, decreasing = TRUE), ]

comparison %>%
  kbl(
```

```

caption = "Table 4.2. Model Comparison Summary",
booktabs = TRUE,
digits = 4,
col.names = c("Model", "R2", "Adj. R2", "AIC", "BIC", "No. Predictors"),
align = "lrrrrr",
row.names = FALSE
) %>%
kable_styling(
  latex_options = c("striped", "hold_position", "scale_down"),
  font_size = 9,
  full_width = FALSE
) %>%
row_spec(0, bold = TRUE, color = "white", background = "black") %>%
row_spec(1, bold = TRUE)

```

Appendix P: Final Model Definition

```

final_model <- model_full
final_model_summary <- summary(final_model)
final_anova <- anova(final_model)

```

Appendix Q: Residual Diagnostic Plots

```

par(mfrow = c(2, 2))
plot(final_model)
par(mfrow = c(1, 1))

```

Appendix R: Normality of Residuals — Q-Q Plot and Histogram

```

par(mfrow = c(1, 2))

qqnorm(residuals(final_model), main = "Normal Q-Q Plot of Residuals")
qqline(residuals(final_model), col = "red", lwd = 2)

hist(residuals(final_model), breaks = 50, col = "steelblue",
      main = "Histogram of Residuals", xlab = "Residuals")
curve(dnorm(x,
            mean = mean(residuals(final_model)),

```

```

        sd    = sd(residuals(final_model)) * length(residuals(final_model)) *
        diff(hist(residuals(final_model), breaks = 50, plot = FALSE)$breaks)[1],
        add = TRUE, col = "red", lwd = 2)

par(mfrow = c(1, 1))

```

Appendix S: Normality Tests — Shapiro-Wilk and Kolmogorov-Smirnov

```

set.seed(33)
resid_sample  <- sample(residuals(final_model), min(5000, length(residuals(final_model)
shapiro_result <- shapiro.test(resid_sample)
ks_result      <- ks.test(residuals(final_model), "pnorm",
                        mean = mean(residuals(final_model)),
                        sd    = sd(residuals(final_model)))

```

Appendix T: Homoscedasticity — Residuals vs. Fitted and Scale-Location Plots

```

par(mfrow = c(1, 2))

plot(fitted(final_model), residuals(final_model),
     main = "Residuals vs Fitted", xlab = "Fitted Values", ylab = "Residuals",
     pch = 20, col = rgb(0, 0, 1, 0.3))
abline(h = 0, col = "red", lwd = 2)
lines(lowess(fitted(final_model), residuals(final_model)), col = "orange", lwd = 2)

plot(fitted(final_model), sqrt(abs(residuals(final_model))),
     main = "Scale-Location", xlab = "Fitted Values",
     ylab = expression(sqrt("|Residuals|")),
     pch = 20, col = rgb(0, 0, 1, 0.3))
lines(lowess(fitted(final_model), sqrt(abs(residuals(final_model)))),
     col = "red", lwd = 2)

par(mfrow = c(1, 1))

```

Appendix U: Homoscedasticity — Breusch-Pagan Test

```
bp_result <- bptest(final_model)
```

Appendix V: Independence of Errors — Durbin-Watson Test

```
dw_result <- dwtest(final_model)
```

Appendix W: Multicollinearity — VIF Table for Final Model

```
vif_vals <- vif(final_model)

vif_df <- as.data.frame(vif_vals)
vif_df$Variable <- rownames(vif_df)

vif_display <- data.frame(
  Variable = rownames(vif_df),
  GVIF     = round(vif_df[, 1], 4),
  Df       = vif_df[, 2],
  `Adj. GVIF` = round(vif_df[, 3], 4),
  check.names = FALSE
)

vif_display %>%
  kbl(
    caption = "Table 4.3. Variance Inflation Factors (Final Model)",
    booktabs = TRUE,
    row.names = FALSE,
    align     = "lrrr"
  ) %>%
  kable_styling(
    latex_options = c("striped", "hold_position"),
    font_size     = 9,
    full_width    = FALSE
  ) %>%
  row_spec(0, bold = TRUE, color = "white", background = "black") %>%
  row_spec(which(vif_display$GVIF > 10), bold = TRUE, color = "red")
```

Appendix X: Influential Observations — Cook's Distance and Leverage

```
cooks_d <- cooks.distance(final_model)
threshold <- 4 / nrow(data_clean)

influential <- which(cooks_d > threshold)

plot(cooks_d, type = "h",
     main = "Cook's Distance",
     ylab = "Cook's Distance", xlab = "Observation Index",
     col = ifelse(cooks_d > threshold, "red", "steelblue"))
abline(h = threshold, col = "red", lty = 2)
legend("topright", legend = paste0("Threshold = 4/n = ", round(threshold, 5)),
     col = "red", lty = 2)
```

```
hat_vals <- hatvalues(final_model)
p <- length(coef(final_model))
n <- nrow(data_clean)
lev_threshold <- 2 * p / n

high_leverage <- which(hat_vals > lev_threshold)
```

Appendix Y: Diagnostic Summary Table

```
diag_table <- data.frame(
  Test = c(
    "Shapiro-Wilk (Normality)",
    "Kolmogorov-Smirnov (Normality)",
    "Breusch-Pagan (Homoscedasticity)",
    "Durbin-Watson (Independence)",
    "VIF Max (Multicollinearity)",
    "Cook's Distance (Influential Obs.)"
  ),
  Result = c(
    paste0("W = ", round(shapiro_result$statistic, 4), ", p = ", round(shapiro_result$p.value, 4)),
    paste0("D = ", round(ks_result$statistic, 4), ", p = ", round(ks_result$p.value, 4)),
    paste0("BP = ", round(bp_result$statistic, 4), ", p = ", round(bp_result$p.value, 4)),
    paste0("DW = ", round(dw_result$statistic, 4), ", p = ", round(dw_result$p.value, 4)),
    paste0("Max GVIF = ", round(max(vif_vals[,1]), 2), " (", round(max(vif_vals[,3]), 4), "%)",
    paste0(length(influential), " flagged (", round(length(influential)/n*100, 1), "% of")
  )
)
```

```

),
Status = c(
  ifelse(shapiro_result$p.value > 0.05, "PASSED", "VIOLATED*"),
  ifelse(ks_result$p.value > 0.05, "PASSED", "VIOLATED"),
  ifelse(bp_result$p.value > 0.05, "PASSED", "VIOLATED"),
  ifelse(dw_result$statistic > 1.5 & dw_result$statistic < 2.5, "PASSED", "CHECK"),
  ifelse(max(vif_vals[,3]) < 5, "PASSED", "VIOLATED*"),
  ifelse(length(influential)/n < 0.05, "REVIEW", "ACCEPTABLE")
),
stringsAsFactors = FALSE
)

diag_table %>%
  kbl(
    caption = "Table 4.4. Diagnostic Summary",
    booktabs = TRUE,
    row.names = FALSE,
    align = "llc"
  ) %>%
  kable_styling(
    latex_options = c("striped", "hold_position", "scale_down"),
    font_size = 9,
    full_width = FALSE
  ) %>%
  row_spec(0, bold = TRUE, color = "white", background = "black") %>%
  footnote(
    general = "* Violations are technical rather than substantive; see interpretation below",
    general_title = "Note:",
    footnote_as_chunk = TRUE
  )

```